

Survey paper

Video stabilization: A comprehensive survey

Yiming Wang^{a,b}, Qian Huang^{a,b,*}, Chuanxu Jiang^{a,b}, Jiwen Liu^{a,b}, Mingzhou Shang^{a,b}, Zhuang Miao^{c,*}

^a Key Laboratory of Water Big Data Technology of Ministry of Water Resources, Hohai University, Nanjing, China

^b School of Computer and Information, Hohai University, Nanjing, China

^c Command and Control Engineering College, Army Engineering University of PLA, Nanjing, China



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ABSTRACT

Video stabilization is one of the most fundamental and challenging tasks in video processing, which can be widely applied in many areas, such as video surveillance, robotics, unmanned aerial vehicles and smartphones. It is a video enhancement technology that aims to improve original video quality by removing potentially shaky camera motion. Generally, conventional methods can be divided into three types: 2D, 3D and 2.5D models, which mainly have boundedness. In the past decade, deep learning has emerged as a powerful technique for learning feature representations directly from data, leading to significant progress in video stabilization. However, previous surveys mainly focus on conventional methods and lack performance comparison. In this paper, we present a comprehensive survey of video stabilization. Firstly, we briefly review three stages of video stabilization. Then, we deliver the conventional methods and the deep learning-based methods in detail. Furthermore, we pay special attention to video stabilization quality assessment, benchmark datasets, and state-of-the-art performance. Finally, we provide discussions on current challenges and future directions to overcome the limitations of the existing methods and better meet the needs of researchers in this active area.

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1. Introduction

With the rapid development of computers and multimedia in the 21st century, smartphones have been used more frequently in daily life, and wearable devices (Google Glass [1], etc.) have gradually attracted attention. Most of these hand-held devices have camera capabilities, but videos taken with them are suffering from unwanted motions, which appear as the uncontrolled oscillation of the whole frame. This oscillation can result in a blur effect and significantly degrade video quality. According to subjective research, the instability of a video is easily perceived by humans. These instabilities appear as abrupt motion or different frequencies shaky, which can affect visual perception. Moreover, fast movements induce wobble artifacts and geometrical distortions for hand-held devices equipped with a rolling shutter sensor in the captured scene.

Photographers usually use professional photography equipment to solve these issues. For example, photographers use a tripod to stabilize the camera or a dolly to shoot to achieve a better stabi-

lized video. However, shooting equipment is expensive, cumbersome and not widely available for amateurs. As a result, video stabilization has gradually become a burning research topic. Video stabilization employs hardware or software to stabilize unsteady video sequences. Video quality can be improved by reducing or eliminating unwanted oscillations. Stable video can better represent details and is not easy to cause visual fatigue. Video stabilization can be applied to many areas, including video surveillance, unmanned aerial vehicles (UAVs) and smartphones. It can also improve the accuracy and robustness of object detection, tracking and visual simultaneous localization and mapping (SLAM). Video stabilization can also improve the data compression ratio during the video processing.

Video stabilization has gone through three stages: *mechanical*, *optical* and *digital* stabilization. *Mechanical* stabilization employs a device known as a gyroscopic stabilizer, as shown in Fig. 1. The gyro-stabilizer is an invisible tripod embedded in the camera that compensates for the captured video and outputs stabilized videos. However, these devices are heavy and consume more energy, making them unsuitable for power-sensitive and payload-constrained imaging applications, such as hand-held devices.

Optical stabilization is widely used in smartphones such as Samsung, Apple, etc. and serious SLR cameras such as Nikon, Canon, etc., as illustrated in Fig. 2. This stabilization moves the lens group

* Corresponding author at: Key Laboratory of Water Big Data Technology of Ministry of Water Resources, Hohai University, Nanjing, China (Q. Huang).

E-mail addresses: huangqian@hhu.edu.cn (Q. Huang), emiao_beyond@163.com (Z. Miao).



Fig. 1. Mechanical Stabilizer [2].

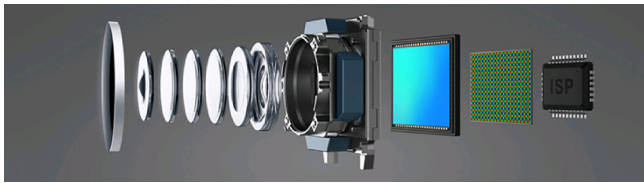


Fig. 2. Optical Stabilizer [3].

on a plane perpendicular to the optical axis to offset image vibration and achieve image stabilization. However, the optical components are relatively expensive and easily affected by light, thus final results may be failed.

Digital stabilization directly estimates the motion vectors of the successive frames without hardware devices, as shown in Fig. 3. The shaky component is separated by filtering and removed by motion compensation. Finally, frames are reconstructed by image warp.

Table 1 compares mechanical, optical and digital stabilization in several aspects.

Compared with the above two video stabilization, digital video stabilization has the following advantages: low cost, small size, light-weight and good stabilization. At the same time, it can perform both online and offline processing depending on the needs. This paper focuses on Digital Video Stabilization (DVS) solutions in particular.

In the field of video stabilization, there are three representative surveys [4–6]. However, literature [4] only focuses on traditional methods, lacking the review of deep learning-based methods. Literature [5] further discusses deep learning-based methods and quality assessment criteria but lacks comprehensive surveys on deep learning-based methods and performance comparison. Literature [6] covers and discusses the traditional, deep learning-based methods but lacks discussions on quality assessment criteria, datasets and the results of the main stabilization methods. Therefore, a more in-depth investigation of video stabilization is indispensable for a better guide of future study. In this paper, a comprehensive

Table 1

Comparison of Video Stabilization Techniques.

	Mechanical	Optical	Digital
Principle	Stabilize whole camera	Adjust optical paths	Use digital processing
Components	Gyroscope	Optical element	None
Process	Online	Online	Offline/Online
Stabilization	Slight	Slight	Significant
Power	High	High	Low
Cost	High	High	Low

investigation of conventional and learning-based methods is provided. Furthermore, we present and discuss the quality assessment, datasets and SOTA methods [7–26].

Compared to the existing surveys [4–6], the main contribution of our work is summarized as follows:

- We provide a comprehensive survey of the most representative methods in video stabilization;
- We present and discuss comprehensive assessment metrics for video stabilization;
- We summarize and discuss the performance of state-of-the-art methods on four widely-used public datasets;
- We further discuss some challenges and future directions for video stabilization tasks.

Rest of the paper is organized as follow: Section 2 reviews 2D, 3D and 2.5D methods in the conventional digital video stabilization. Then, Section 3 details the different components and approaches in deep learning-based video stabilization. Furthermore, Section 4 presents the video stabilization quality assessment methods. Section 5 introduces a summarization of public datasets and the state-of-the-art performances. Also, Section 6 discusses challenges and future directions. Finally, Section 7 concludes this work.

2. Conventional digital video stabilization

Digital video stabilization usually employs a “three-step” approach, illustrated in Fig. 4: (i) motion estimation, (ii) motion compensation, and (iii) image warp. The first step is establishing a motion model and estimating the global motion vectors. In the next step, the motion model is used to remove the shaking to produce stable camera motion. Finally, image warp is performed on the generated stable camera path. Offline stabilization methods tend to process all frames before following the next step separately, whereas the methods for online stabilization generally apply them in sequence for a clip of frames.

Video stabilization is built on the camera motion model and warps the frames to gain visually smooth inter-frame motion. Motion model can be used to approximate camera motion. It is based on geometrical models that describe the dimensionality of



Fig. 3. Digital Video Stabilization.

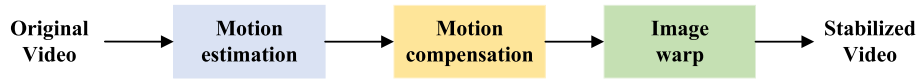


Fig. 4. Diagram of video stabilization algorithm.

motion representation, which represents the physical process of capturing a scene with a hand-held camera. Early methods apply full-frame 2D affine or projective warps to model camera motion. However, 2D approaches cannot model the parallax generated by translational shifts. Therefore, 3D methods are proposed to solve this problem. However, modeling parallax from a video sequence is a time-consuming and difficult process that may fail in many cases. Thus, hybrid methods(2.5D) are proposed to transform 2D motion trajectories to achieve visually plausible views of 3D methods and robustness of 2D methods, but at the cost of distortion and physically-inaccurate results.

According to different motion models, video stabilization techniques are divided into *two-dimensional (2D)*, *three-dimensional (3D)*, and *hybrid methods (2.5D)* between two-dimensional and three-dimensional. The taxonomy of conventional digital video stabilization is shown in Fig. 5.

2.1. 2D video stabilization

The 2D stabilization methods are widely used in commercial software owing to their robustness. This method only describes the camera motion in the two-dimensional plane and chooses the motion model of the successive frames, such as homography or affine transformation. The 2D methods include motion estimation, motion compensation and image warp.

2.1.1. Motion estimation

A critical step of video stabilization is estimating camera motion vectors, which determines the accuracy and robustness of the system. At the same time, it also affects subsequent operations such as motion compensation and image warp. Camera motion vectors are the global motion vectors in the entire image. Global motion esti-

mation is usually based on local motion estimation. The local motion estimation methods can be divided into block matching, optical flow, sub-plane matching, phase correlation and feature point matching.

(1) Block matching

Block matching is a method of matching selected blocks in the adjacent frame according to the metric, such as Mean Absolute Difference(MAD), Normalized Cross-Correlation(NCC) and Mean Square Error(MSE). Table 2 describes some metrics. Rectangle block size is $M \times N$, $f_c(x, y)$ is the pixel value of the current frame at (x, y) , $f_r(x + \Delta d_x, y + \Delta d_y)$ is the pixel value of the reference frame within the search area $(M + 2\Delta d_x) \times (N + 2\Delta d_y)$. In this method, the frame is divided into square blocks of specific dimensions, and a motion search is performed in the current frame to locate and match specific blocks according to the metric. The process of matching blocks between the reference frame and the current frame constitutes local motion vectors. These local motion vectors are used to generate the global motion vectors.

The block matching method has the advantages of simple principles and convenient implementation. However, the block matching method faces two main problems: how to reduce the computational complexity and how to improve the accuracy of the global motion vectors. The common methods are as follows: selecting motion search strategy, distinguishing foreground and background, using different templates, filtering outliers, and optimizing global motion vectors.

① Select motion search strategy

K.-H Tzou and Li et al. [27,28] propose exhaustive search methods for motion estimation. All candidate blocks are tracked in the search area, and the best-estimated candidate block is obtained

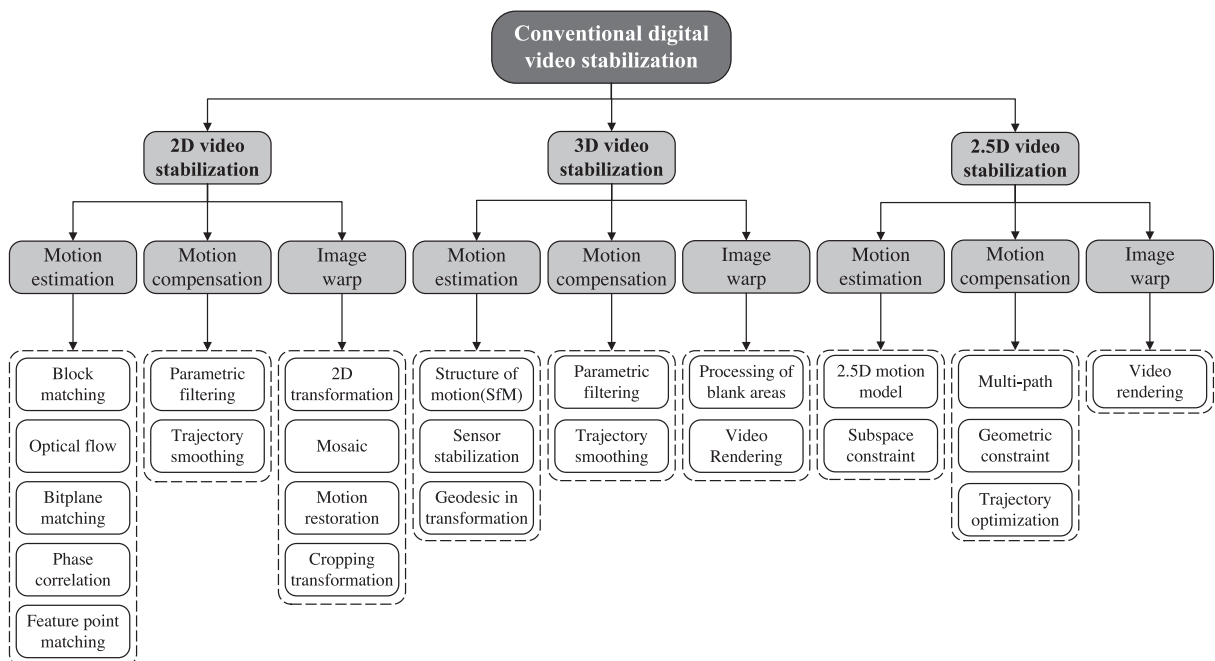


Fig. 5. A taxonomy of conventional digital video stabilization.

Table 2
Metrics for block matching.

Metric	Formula
MAD	$\frac{1}{MN} \sum_{x=1}^M \sum_{y=1}^N f_c(x, y) - f_r(x + \Delta d_x, y + \Delta d_y) $
NCC	$\frac{\sum_{x=1}^M \sum_{y=1}^N f_c(x, y) f_r(x + \Delta d_x, y + \Delta d_y)}{\sqrt{\sum_{x=1}^M \sum_{y=1}^N f_c^2(x, y)} \sqrt{\sum_{x=1}^M \sum_{y=1}^N f_r^2(x + \Delta d_x, y + \Delta d_y)}}$
MSE	$\frac{1}{MN} \sum_{x=1}^M \sum_{y=1}^N [f_c(x, y) - f_r(x + \Delta d_x, y + \Delta d_y)]^2$

according to the metric. The exhaustive search algorithm improves block matching accuracy but also increases computational complexity. To reduce the computational complexity, Jain et al. [29] propose a 2D logarithmic search method, Koga [30] proposes a three-step search method (TSS), Li et al. [31] propose a new three-step search method (NTSS), LM Po et al. [32] propose four-step search (FSS), Zhu et al. [33] propose diamond search (DS) and so on. Wang Y [34] summarizes the above conventional search strategies in the literature.

② Distinguish foreground and background

Vella et al. [35] divide the foreground and background of the video based on experience. Specifically, the foreground is located in the central area of the frame, and the background is close to the frame boundary. The authors calculate the motion vectors of the foreground and background independently, then decide whether to stabilize the foreground or background to improve the accuracy of the block matching. The authors use the weighted square histogram method to estimate the foreground and background motion vectors respectively.

③ Use different templates

In the block matching method, the matching template is often square or rectangle. However, Xu et al. [36] firstly propose circular block matching based on rotation invariant characteristics to mitigate the rotation impact. This method minimizes the computational cost and improves the matching accuracy of the block matching method.

④ Filtering outliers

Due to matching or calculation errors, the obtained local motion vector will have outliers and the results of global motion vectors directly based on them are inaccurate. Therefore, local motion vectors must filter outliers. Battiatto et al. [37] propose a method based on block matching of local motion vectors. Based on the matching and the similarity of local motion vectors, this approach is utilized to perform pre-filtering and filter outliers. The memory filtering is then performed based on the wrong motion vector caused by object motion in the frame. The proposed method operates in real-time and achieves effective experimental results even under challenging conditions (light changes, moving objects, image blur).

⑤ Optimizing global motion vectors

Puglist et al. [38] effectively extract a collection of motion vectors by block matching, then strengthen local motion vectors using an exhaustive search strategy and a projection-based integral error function. It is proposed to apply a vote-based filtering step for least-squares methods to optimize the estimation of global motion vectors.

(2) Optical flow

The optical flow (see Fig. 6) is firstly proposed by Gibson [39] in 1950. It is a method to infer the object motion information by detecting the changes of pixels in the frame and the relationship between successive frames in the temporal and spatial domains.

In 1981, Lucas and Kanade [41] propose a widely used differential method for optical flow. It is assumed that the optical flow in the neighbourhood of pixels is a constant and the least-squares method is used to solve the optical flow equation for all pixels in the neighbourhood, which provides a foundation for the use of optical flow in video stabilization. In 1994, Barron et al. [42] summarize and compare several methods including differential methods, region-based matching, capability-based and phase-based methods. The experimental results show that Lucas and Kanade method is the most reliable. Chang et al. [43] estimate camera motion by calculating the optical flow between consecutive frames and fitting a simplified affine motion model to the optical flow vectors with the least-squares method. Jean-Yves Bouguet et al. [44] propose a pyramid iterative Lucas-Kanade method for optical flow, which is an improved Lucas-Kanade algorithm for affine transformation. Cai et al. [45] adopt the pyramid-based iterative Lucas-Kanade method for optical flow and propose to estimate the camera motion from the histogram of local motion. The study [46] shows that most optical flow may fail when there are large moving objects in the scene. Simultaneously, the authors demonstrate that delta optical flow can be used to improve the robustness of video stabilization. In 2011, Baker et al. [47] give a brief overview and evaluation of the most representative optical flow. Liu et al. [48] replace feature trajectories with optical flow to deal with spatially-variant motion. They also propose a novel motion model, SteadyFlow. A SteadyFlow is a specific optical flow that enforces strong spatial coherence, allowing the smooth feature trajectory to be replaced by a smooth pixel contour, which is a motion vector collected at the same pixel position over time. It can avoid fragile feature tracking in video stabilization system. Experiments show that SteadyFlow method is effective. Kurnianggoro et al. [49] propose an optical flow-based method to detect moving objects in videos. The motion parameters are estimated using motion estimation and the dense optical flow is subjected to the threshold judgment to obtain the area of the moving object. With the development of deep learning, more learning-based methods are being proposed in this field. Yu et al. [19] present a novel Selfie stabilization method that uses optical flow to track background pixels and 3D human face data to learn about foreground motion. The method is robust under different selfie scenes. Zhai et al. [50] make a summarization of the optical flow, the learning-based optical flow will be introduced in the Section 3.

(3) Bitplane matching

Most algorithms for estimating the global motion vectors have high computational complexity while maintaining motion estimation accuracy. Bitplane matching can be realized using only Boolean functions, significantly reducing computational complexity.

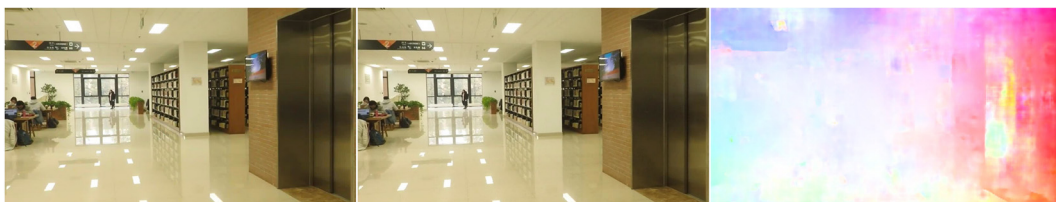


Fig. 6. Optical flow estimated by PWC-Net [40].

In 1998, Ko et al. [51] propose a bit-plane matching method, which performs motion vector matching by XOR matching of 1-bit planes extracted from video sequences. The experimental results show that the algorithm performance is comparable to the full search block matching algorithm, but the method significantly reduces computational complexity. To further reduce computational complexity, the authors [52] propose a new motion estimation and compensation method based on grayscale coding. The proposed method uses grayscale coding bit-plane in a video sequence to perform fast binary matching to determine local motion vectors from sub-images of the frame. A multiplication-free 1-bit-plane transform (MF1BT) method is proposed in literature [53], which reduces the computational cost by utilizing a novel multiplication-free method to obtain the bit-plane. In literature [54], the authors introduce a 2-bit plane transform (2BT) method to improve the performance of the 1-bit plane transform, thus increasing computation complexity. In the literature [55], a constrained 1-bit-plane transform (C-1BT) method is proposed, which introduces a constraint mask so that only reliable pixels of the 1-bit plane transform are used in the matching process. Compared with (1BT) and (2BT) methods, the matching accuracy of (C-1BT) method is higher. Literature [56] employs a partial distortion search method combined with a sparse search point method with a C-1BT to reduce the computational complexity at the expense of performance. In the literature [57], an earlier scheme is combined with an MF1BT-based method to reduce computational complexity.

(4) Phase correlation

A frequency-domain method based on the Fourier transform is phase correlation. It estimates the relative offset between two successive images using a fast frequency-domain method. An image with a peak intensity appears at the best matching position. The advantage of the method is that it does not require a threshold and produces better effects than the grayscale detection method. At the same time, phase correlation is not affected by lighting. It has anti-noise capabilities, good stability and can detect multiple object motion. The disadvantage is that it has a poor effect and exponentially increases the computational complexity in some cases, such as blurred and larger images.

Erurk et al. [58] propose a Fourier frequency domain filtering method based on absolute frame displacement. By estimating the absolute displacement of the accumulated frame, the global

motion vector of the frame is obtained. In 2003, the author [59] proposes a global motion estimation method based on sub-image phase correlation and motion compensation based on Kalman filtering. In the current frame, four sub-images of 64×64 pixels are chosen and then local motion vectors are estimated from each sub-image of the previous frame by using motion search based on phase correlation. Finally, some strategies are chosen based on the local motion vectors, such as weighted averaging or assigning the local motion vector with the most significant peak amplitude as the global inter-frame motion vector. Sanjeev et al. [60] propose a FourierMellin domain phase correlation method. The rough estimation is performed firstly in the motion estimation and then refined using a sparse motion vector field. The obtained results are very robust to outliers such as foreground objects and flat regions. Ayhan et al. [61] implement Erurk's algorithm in Texas instruments' DaVinci DSP DM6437 kit, and Maik et al. [62] combine the local phase correlation method and feature analysis for video stabilization.

(5) Feature point matching

Feature detection(see Fig. 7) is a critical component of feature matching. It is used to detect extracted image information and determine whether the detected points belong to images features. It directly simplifies the expression of high-dimensional image data. These features are associated with individual points, continuous curves or continuous regions. Common feature point detection is divided into edge detection, corner detection, ridge detection and blob detection.

① edge detection

Edge detection is a technique used in image processing to determine the boundaries of objects within images. In video stabilization, the commonly used edge detection methods include Laplace [64], Roberts [65], Prewitt [66], Sobel [67], Canny [68], Laplacian of Gaussian (LoG) [69], SUSAN [70], Difference of Gaussian (DoG) [71] operator. Currently, the Canny operator is the most widely used edge detection method.

② corner detection

In images, a corner point is usually defined as the intersection of two sides. A corner can also be defined as a point with two dominant and opposing edge directions in its local neighbourhood. The widely used corner detection in video stabilization includes Moravec [72], Harris [73], Shi-Tomarsi [74], FAST [75] and KLT [76] algorithms. The Moravec [72] is one of the earliest corner detection



Fig. 7. Feature point detection and matching by GMS [63].

algorithms, but it does not solve the noise problem and has a simple response to edges. Harris et al. [73] propose the Harris corner detection algorithm and improve the Moravec algorithm in 1988. Instead of using shift patches every 45 degrees, the Harris method directly refers to the direction. It considers the difference of the corners, and the experiment has proved to distinguish edges and corners more accurately. Shi-Tomasi corner detection [74] later assumes that each pixel function value is generally smooth and stable. The corner points are found by computing the minimum of the two feature values. This method is also known as Kanade-Tomasi angle detection. In 1997, Smith and Brady [77] propose the SUSAN corner detection method. The detection is performed by comparing the similarity of the concentricity in the pixel field. The method is accurate and anti-noise. In 2006, Rosten and Drummond [75] propose the FAST corner detection algorithm. The method is highly efficient in detection and can achieve excellent performance in terms of computational time and resources. Kanada-Lucas-Tomasi Tracking corner algorithm, also known as the LK algorithm, is the classic corner algorithm. The KLT algorithm uses spatial intensity information to guide the search for the best match location. However, the KLT method may fail when the appearance around key points varies greatly due to occlusion, illumination, viewpoint change and motion blur.

③ ridge detection

Ridge detection is a technique for locating ridges in an image. The Hough transform is a ridge detection method designed initially to detect analytically defined shapes (e.g. lines, circles, ellipses, etc.). Dana H. Ballard [78] proposes the generalized Hough transform method in 1981, which is adjusted according to the template matching principle based on the Hough transform and can detect any shapes.

④ blob detection

Blob detection detects areas in an image that are qualitatively different from surrounding areas, such as brightness or color. The LoG, DoG and Hessian variance determinant (Determinant of the Hessian, DoH) methods are widely used for speckle detection. In the video stabilization, the DoG method is used for the Scale-Invariant Feature Transform (SIFT) algorithm [79]. A simplified form of the DoH method adopts Haar wavelets to calculate the scale-normalized determinant of the Hessian matrix for the Speeded Up Robust Features (SURF) algorithm. Among them, the SIFT is proposed by Lowe in 1999 and summarized in 2004. The algorithm is invariant to uniform scaling, orientation, lighting changes and affine-distorted parts. In 2006, Bay et al. [80] propose the SURF algorithm. The SURF is based on the SIFT, but the detailed process is slightly different. The PCA-SIFT algorithm, proposed by Ke et al. [81], uses principal component analysis to reduce memory usage and speed up pairing.

Following the detection of feature points, feature matching is performed. Abdullah et al. [82] use the Harris method and the sum of squared differences as the metrics to obtain rough matching results. To obtain more accurate matching results, the Random Sample Consensus (RANSAC) algorithm [83] is used to eliminate incorrect matching pairs and improve the geometric transformation. Xu et al. [84] use the FAST method for feature extraction and propose rotational binary robust independent elementary features (BRIEF) for adjacent frame matching. The directional FAST method and the combination of rotated BRIEF is known as Oriented FAST and Rotated Brief (ORB), and it is very effective in both feature detection and matching. Wang et al. [85] calculate the motion vector by using the KLT method. Chen et al. [86] propose a method based on line characteristics. The algorithm estimates the motion vector using Hough transform. The whole line feature method is used to eliminate the influence of local motion. Battiatto et al. [87] propose a method based on frame extraction and scale-invariant features tracking. SIFT features are extracted from

frames, and their trajectories are then evaluated to estimate inter-frame motion. The iterative least-squares method avoids estimation errors and improves video stability. Lee et al. [88], Yang et al. [89], Chen et al. [90], and Fang et al. [91] all use the SIFT method in their algorithms. Despite the SIFT remarkable success in video stabilization, its computational complexity is relative high, so Sultan et al. [92] propose to adopt the SURF method as a method for feature extraction and improving the selection process of extracted features for global motion.

2.1.2. Motion compensation

The global motion vector is obtained during motion estimation. In the motion compensation, the global motion vectors are compensated so that the active movement and shakes are separated to produce smooth motion trajectories, which is the essence of stabilization. There are two types of motion compensation methods: parametric filtering methods and trajectory smoothing methods.

(1) A smoothing method based on parametric filtering

① Mean filtering

Mean filtering is also known as linear filtering. It is a simple sliding-window spatial filter that replaces the center value with the average values in the window. Kim-Loan et al. [93] propose a moving simplified mean filter to achieve smooth motion. Lim et al. [94] propose a mean filtering method for detecting active movement and shakes. Mean filtering is used to eliminate shaking and stabilize videos.

② Gaussian filtering

In image processing, Gaussian filtering combines the frequency domain and time domain. It is a low-pass filter to reduce high-frequency and smooth the frame. Rawat et al. [95] propose to use Gaussian filtering to filter the global motion vectors. The Gaussian smoothing method is similar to the mean filter, which will cause blur images and the smoothness is determined by the standard deviation of the Gaussian function. Fang et al. [91] adopt Gaussian filtering to preserve the active movement of the camera and lose small pixel information.

③ Kalman filtering

Kalman filter, also known as linear quadratic estimation (LQE), is a recursive filter. It iteratively processes a stream of noisy input data to produce the statistically best estimate of the underlying system state. Literature [96] uses a Kalman filter to smooth camera motion. The extended Kalman Filter (EKF) method is proposed by [97] for distinguishing active and inactive camera motions. This method produces very accurate results in most cases, but it needs to adjust the motion model to match the type of camera motion. Yu et al. [98] use Kalman filtering and the least-squares fitting algorithm to remove shaky motion and keep the camera stable.

④ Particle filtering

The particle filter is a recursive filter that employs the Monte Carlo method [99]. It is a generalized Kalman filter based on linear state-space and Gaussian distributed noise. The particle filter state-space model can be nonlinear, and the noise distribution can be of any type. Yang et al. [100] propose particle filtering for motion smoothing, which overcomes the limitations of Kalman filters. The expected motion parameters constantly change in the particle filter, which aids in distinguishing and maintaining camera movement expectations. Zhu et al. [101] use particle filtering to obtain smooth motion vectors from the original motion curves.

(2) A fitting method based on trajectory smoothing

Different from parameter filtering, the fitting method imitates the characteristics of professional camera motion paths and decomposes stable camera paths into static paths, uniform motion

paths, uniform acceleration paths and the path between uniform speed and uniform acceleration. Most parametric filtering methods can only suppress high-frequency motion and cannot eliminate low-frequency shakes or distortion that occurs during hand-held device shooting. Gleicher and Liu [102] are the first to use cinematography optimization to fit the camera motion path. A camera path with constraint is proposed. Grundmann et al. [8] use it to approximate the input path by building multiple sparse motion models with saliency, blur and clipping window constraints. The optimal stable camera path minimizes the function $\mathcal{C}(P)$.

$$\mathcal{C}(P) = w_1 \|D(P)\|_1 + w_2 \|D^2(P)\|_1 + w_3 \|D^3(P)\|_1 \quad (1)$$

where $D(P)$ represents a constant path, $D^2(P)$ is a constant velocity path, $D^3(P)$ is a constant acceleration path and w_1 , w_2 and w_3 are the weights. The optimization is designed to minimize $D(P)$, $D^2(P)$ and $D^3(P)$ separately.

Grundmann et al. [103] further use a homography-array-based motion model to deal with rolling shutter effects. These two techniques have been integrated into Google YouTube [104] and perform well in many consumer videos. Qu et al. [105] propose a new hybrid L1-L2 optimized method based on the research of [8] that aims to eliminate unnecessary camera motion in the path fitting stage while preserving the original video information to the greatest extent. In order to further improve video stabilization, Zhang et al. [106] use the L1 camera path optimization framework to solve the low-level and high-level constraints on the camera path through optimization to achieve the goal of motion compensation. Zhao et al. [107] also work on a trajectory matrix composed of many local trajectory matrices extracted from long trajectories to smooth the frames. Liu et al. [48] propose to replace feature trajectory smoothing with pixel contour smoothing, which has several advantages over traditional feature trajectory smoothing. Inspired by [48], Liu et al. [10] propose a computationally efficient motion model, MeshFlow. It is a sparse and spatially smooth motion field, with its motion vector only located at the mesh's vertices. The video with spatially varying motion can be stabilized by smoothing the vertex contour and the motion vectors at the mesh vertices over time. MeshFlow is computationally more efficient than SteadyFlow. However, the method cannot handle videos with larger near-range foreground objects. Liu et al. [108] propose a new motion model called CodingFlow, performed from the motion estimation during the video coding procedure. They regularize these motion vectors to produce a spatially smoothed motion field at each frame. It is designed to optimize motion compensation. Based on this, a grid-based 2D method is proposed to achieve video stabilization. Zhang et al. [109] propose a novel method to exclude distractors in the background for L1 path optimization.

2.1.3. Image warp

After the motion estimation and motion compensation, a new and smooth motion chain is obtained. According to the motion model, the image patching operation deforms the original frame to produce a stable video containing the missing part (blank edge). Image warp is a technique that combines image stitching, super-resolution, fusion, and image inpainting.

Two-dimensional image warp mainly includes 2D transformation, mosaic, motion inpainting and cropping transformation. The 2D transformation method is a relatively simple implementation. It performs a one-to-one transformation on the input and output frames. In this method, blank areas will appear due to the loss of image pixels in the motion compensation when the video is output, as shown in Fig. 8. In order to solve this situation, Litvin et al. [96] use the Mosaic method to fill the blank area through transformation estimation and the pixel value of successive frames

by using the time correlation in the image warp. However, mosaic method may cause artifacts in non-planar scenes and moving object scene areas. Given this situation, Matsushita et al. [110] propose a classical image warp method based on motion inpainting, incorporating the concept of local motion estimation into image warp. It is propagated to the blank area that cannot be calculated directly based on the adjacent frames. The propagated motion field can naturally fill the missing image area, which is also effective for non-planar scenes and moving object scene areas. Grundmann et al. [8] adopt cropping transformation, which only retains the content within the cropping window to eliminate the influence of blank areas. Both the mosaic and the motion inpainting methods can produce full-frame, but the computational complexity is high, and the splicing area of each frame will transition unnaturally. The cropping transformation is simple to implement and the transition of each frame is continuous and natural, but some frame information will be lost.

2.2. 3D video stabilization

In general, the 2D video stabilization cannot solve the parallax problem caused by the translational motion in the viewpoint. In the face of the scene with parallax, the 2D video stabilization methods perform poorly.

Unlike the 2D method, 3D video stabilization considers camera motion information in the 3D space. It estimates motion information from the 2D video sequence and reconstructs 3D motion information. The structure from motion (SfM) algorithm is mainly used. Then recovered 3D camera motion information is compensated to separate the active camera motion from the jitter, resulting in a smooth camera motion trajectory. Finally, image warp is performed to produce a stabilized video. The 3D method uses the same processing steps as the 2D method: motion estimation, motion compensation and image warp.

2.2.1. Motion estimation

In the 3D video stabilization algorithm, motion estimation is primarily based on the structure of motion (SfM) to estimate the 3D motion information. The methods must firstly match the feature points of successive frames and then use the matched feature point pairs to reconstruct the 3D motion information. The motion estimation methods of 3D video stabilization can be divided into Structure of Motion (SfM) algorithms, sensor stabilization and geodesic in transformation algorithms.

(1) Structure of motion(SfM)

Liu et al. [111] develop the first successful 3D video stabilization system. The authors firstly adopt the Structure of Motion (SfM) to recover 3D camera motion and sparse 3D point cloud to obtain stable camera paths by fitting a linear path. Finally, the least-squares method is used to optimize. Ren et al. [112] propose a stabilization method using a 3D rotational motion model. The feature points are matched using the optical flow, and then the rotation matrix between frames is calculated using the Spherical Motion Structure algorithm. The motion trajectory of the camera can be represented by a sequence of 3D rotation matrices. The authors take the manifold structure of the rotation matrix sequence and smooth the 3D rotation matrix sequence based on the geodesic distances on the nonlinear manifold. However, the SfM algorithm is slow and not robust enough to stabilize videos with less texture or parallax. Zhou et al. [113] propose a plane-based content-preserving warps method to estimate a single homography for regions belonging to the same plane while borrowing the results of Liu et al. [111] in non-planar regions and segmentation information can be efficiently obtained to stabilize video. Liu et al. [114] propose

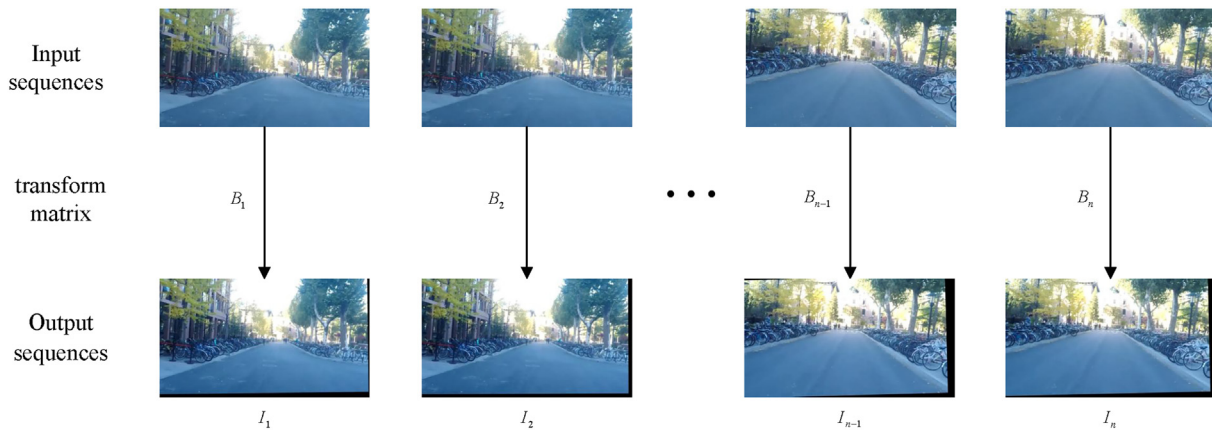


Fig. 8. 2D transformation.

a method that chooses different methods according to depth variation, making it the first video system to combine the benefits of both 2D and 3D methods. Kopf et al. [115] propose a hybrid 3D-2D method for 360° video stabilization. The 3D analysis is taken to estimate the rotation, and 2D optimization is used to smooth the feature point trajectories, which can handle small translational jitter, parallax and rolling shutter wobble.

(2) Sensor stabilization

Because the structure-of-motion (SfM) has high computational complexity, Smith et al. [116] use light field video cameras to avoid 3D motion model reconstruction by directly computing a sequence of relative positions between camera arrays. In order to produce the videos that contain sufficient depth motion information, Liu et al. [117] use a depth sensor (i.e., a Kinect camera) to capture the shaking video, which is used for each frame to provide depth motion information and is used to reconstruct the camera 3D motion information. However, the applicability of this method is limited because the vast majority of consumer videos currently do not include depth information. Jia et al. [118] employ a gyroscope to track rotational motion and calculate spatial rotation and translation. The camera path is then completely reconstructed in 3D space, and the Riemannian geometry method is used to smooth the rotation information in spatial motion. Chu et al. [119] use a gyroscope inside the mobile device to calculate the 3D motion information to improve the visual experience. Experiments show that even when the light is low and there are many foreground objects moving, the computational complexity remains low and the stabilization effect is good. Furthermore, Bell et al. [120] propose a gyroscope-based method for hand-held devices to correct the rolling-shutter and stabilize the video. The method utilizes gyroscope data to filter the camera angular velocity, creating the impression of a video shot on a tripod and outperforming previous gyroscope-based methods. However, the method is less robust and cannot run in real-time for parallax scenes. Guo et al. [121] propose a method to stabilize shaky stereoscopic videos by utilizing MeshFlow[10] in motion estimation, joint disparity and stability mesh warp(JDSW) [122] to render the stabilized stereoscopic videos.

(3) Geodesic in transformation

Zacur et al. [123] introduce the concept of computing Riemann geodesics on Lie groups embedded in transform spaces. Zhang et al. [124] propose a new stabilization formulation based on Lie groups embedded in transform spaces, with optimal motion paths projected as geodesics. Finally, motion smoothing is achieved by geometric interpolation. Huang et al. [125] propose

a method for encoding shaky video that integrates efficient video stabilization. A light-weight and high-quality geodesic video stabilization method is employed in video coding with the graceful integration of the stabilization and encoding to achieve better performance.

2.2.2. Motion compensation

The motion compensation of the 3D method differs from the 2D method. The difference is that reconstructed 3D motion information is much more complicated than 2D motion information. It contains not only the camera motion information but also its orientation information. As a result, the camera motion and orientation information should be distinguished for compensation. Motion compensation is classified into two types: local linear and view warp.

(1) Local linear

Because 3D motion orientation information is nonlinear, most compensation schemes cannot be applied to 3D stabilization orientation information. In response to this situation, Lee et al. [126] propose a motion compensation mechanism that converts orientation information into a vector space representation, filters it in the vector space, and finally converts the result back to orientation information. This method is categorized as a local linear method. Liu et al. [111] divide 3D camera motion information into translation and rotation components, which are then smoothed respectively.

(2) View warp

Zhang et al. [127] use a view warp method in the motion compensation. To produce smooth motion information, a planar structure is used to construct a geometric scene, and a constrained warping error function is used to achieve visually reasonable results.

2.2.3. Image warp

After the motion compensation of 3D method, it is necessary to perform image warp on the smooth motion trajectory to output stable videos. The common problems of image warp in 3D methods can be divided into two categories: processing of blank areas and rendering of video sequences.

(1) Processing of blank areas

The image warp of the 3D method is different from the 2D method because the 3D motion information contains the parallax problem, which the 2D video processing algorithm cannot handle. As a result, the 2D image warp method is no longer applicable to the 3D motion trajectory.

To address this problem, Liu et al. [111] propose a content-preserving warps (CPW) algorithm, which is a one-to-one transformation of input and output frames. Unlike the 2D image warp method, it divides the frame into many uniform grids and calculates the transformation matrix for each grid. A shape-preserving image warps method is used in the content-preserving warps method. The deformation of the local shape and the adjustment of the content-aware image are minimized while warping the image, and the content is preserved as much as possible when changing the image size. However, if the scene contains large textureless areas, such as the ground, building surfaces or indoor walls, and there are few feature tracks, the content-based retention method is no longer effective. Facing this situation, Zhou et al. [113] improve Liu's method [111]. Firstly, the frame is divided into planar and non-planar parts. The non-planar part of the frame is transformed based on content-preserving, and the planar part is transformed by homography. This method has been demonstrated to produce a stable video that corresponds to human eye perception.

(2) Rendering problems of video sequences

The 3D image warp can be regarded as the video sequence's rendering process. Unlike 2D image warp, the video sequence's rendering method is not limited to the one-to-one correspondence between input and output frames, as shown in Fig. 9. Its working principle is to convert input sequences to output video sequences using rendering system interpolation, such as deblurring, stitching and fusion. Image-based rendering and optical flow-based rendering methods are used to render video sequences.

① Image-based rendering

Zitnick et al. [128] propose an image segmentation-based stereo algorithm that creates Markov random fields for video frames and uses recurrent belief propagation to update disparity estimation for each segment. The Middlebury stereo dataset [129] is used to validate the algorithm. However, the proposed stereo rendering algorithm may lead to blurring or ghosting artifacts. To solve this situation, Bhat et al. [130] propose an image-based rendering (IBR) algorithm. It is forced to observe the change in scene point radiance overtime to restore the reconstructed video's high-frequency details and remove blur.

② Optical flow-based rendering

Unlike image-based rendering, optical flow field rendering is based on video sequences captured by different cameras simultaneously, rather than video sequences taken by the same cam-

era at different times. Barndon et al. [116] employ a temporal and spatial optimization approach to directly compute the desired virtual camera path relative to the input camera array. Image features are projected into the virtual output viewpoint by using the depth motion information, and the acceleration in the image plane measures the smoothness of the features in the virtual viewpoint. Finally, a stable video is produced.

The video sequence rendering method can get video information for whole frame without losing pixels. The visual transition is continuous and natural, but the computational complexity is usually high.

2.3. 2.5D video stabilization

To address the issue that 2D video stabilization cannot deal with parallax well, 3D video stabilization is developed to solve parallax problem in the scene. However, the 3D model is too sensitive to the noise and change of the scene, and its robustness is weak.

The 2.5D video stabilization is a compromise solution for 2D and 3D video stabilization, it contains motion estimation, motion compensation and image warp.

2.3.1. Motion estimation

In video stabilization, a 2.5D motion model estimates the camera motion of successive frames. It relaxes all 3D reconstruction to partial 3D information reconstruction, which reduces computational complexity and improves robustness. The motion estimation methods are 2.5D motion model estimation, subspace constraint, and trajectory optimization.

(1) 2.5D motion model estimation

Jin et al. [131] propose a 2.5D inter-frame motion model to eliminate shaky video with parallax in the scene. This method implementation consists of four modules: pyramid-based motion detection, 2.5D motion parameter estimation, inertial motion, and motion compensation. This system smooths unwanted jitters in video sequences. Shen et al. [132] present a 2.5D approach to 360° panorama video stabilization. It separates rotation from other motions and handles them respectively.

(2) Subspace Constraint

Liu et al. [7] propose a classical 2.5D video stabilization. Since 3D reconstruction is complicated, Liu et al. [7] directly track the trajectory of features for smoothing. The authors use KLT

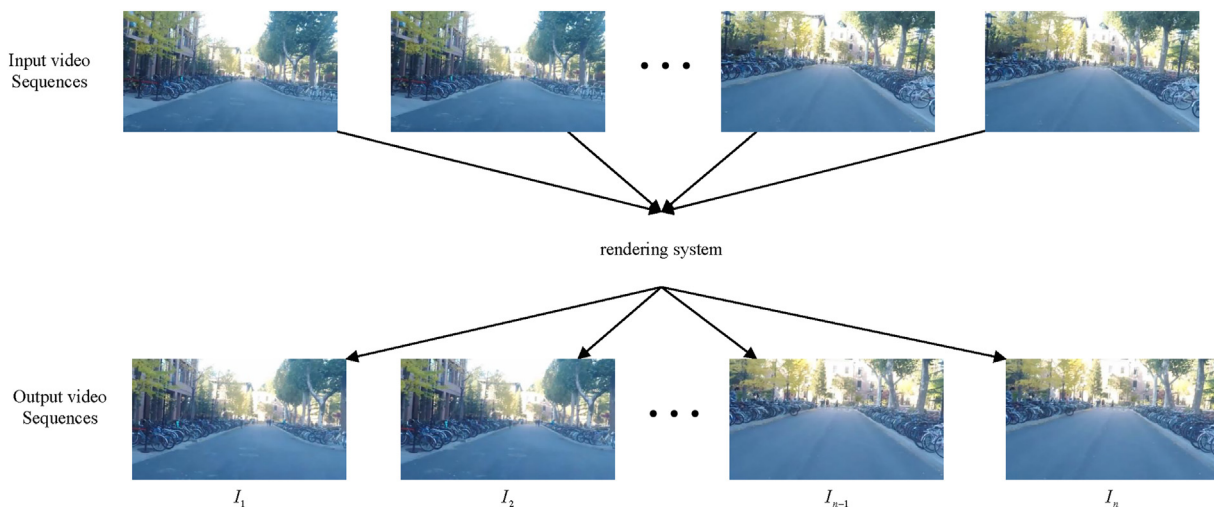


Fig. 9. Rendering systems.

feature point detection [74,75] to calculate a set of sparse 2D feature trajectories. The motion factor decomposition is then carried out. The 2D feature trajectory is limited to a low dimensional subspace, which is effectively fitted to the approximate input motion in a subspace to obtain the camera motion information. This method achieves a quality similar to the 3D reconstruction method and reduces the requirement of long feature trajectory tracking. It has been integrated into the “warp stabilizer” feature in Adobe Premiere Pro 2020.

Given the complementary benefits of 2D and 3D methods, the 2.5D method does not explicitly reconstruct the 3D structure of the camera path but instead employs some spatial relations as path smoothing constraints. For example, Goldstein and Fattal [133] exploit the “epipolar geometry” technique for intermediate scene modelling, known as projection reconstruction. The model explains the internal geometric relationship between the views of general uncalibrated cameras and avoids 3D reconstruction. Experiments show that this method works for any scene with variable depth. Matthias et al. [103] propose the idea of mixed homography, which is fitted to the matching after feature extraction and RANSAC purification to obtain global motion parameters. Wang et al. [134] express each track as a Bezier curve and maintain the spatial relationship between tracks by retaining the original offset of adjacent curves. Video stabilization is defined in this method as a time-space optimization problem that can find smooth features while avoiding visual distortion.

(3) Trajectory optimization

Koh et al. [135] produce virtual trajectories by using a low-rank matrix completion scheme to augment incomplete trajectories when the number of estimated trajectories is insufficient. The proposed algorithm reconstructs more stable videos in extreme scenes, such as large moving objects, non-planar depth variations and rolling-shutter distortions. Zhang et al. [136] propose a new method from the perspective of global optimization. The trajectory assists the local similarity transformation in the motion estimation to produce stable video. However, the video will still be distorted due to the drastic depth change. When 3D reconstruction is difficult or long feature track video cannot be obtained, these methods integrate some 3D structures in the 2D parameter transformation, reducing the computational complexity, and typically providing good stabilization results. However, the 2.5D video stabilization may not eliminate more severe or complex shaking.

2.3.2. Motion compensation

The motion compensation of the 2.5D methods works similarly to the 2D methods. Unwanted jitter is removed from the estimated global motion parameters. Motion compensation is carried out to generate a smooth and stable motion trajectory. In motion compensation, it can also be divided into three types: multi-path method, geometric constraint method and global optimization method.

(1) Multi-path

Gleicher et al. [137] divide the video into multiple segments according to the principle of film shooting, classify each segment and fit the curve fitting for different scenes to produce smooth motion trajectories. Liu et al. [7] propose a multi-trajectories optimization method for motion information in a low-dimensional linear space. It calculates global motion trajectories for different types of video regions and then performs a path fitting method based on trajectory smoothing on these motion trajectories. Simultaneously, constraints are added to ensure that the processed paths are similar. The algorithm can effectively handle the parallax problem in the video.

(2) Geometric constraint

Goldstein and Fattal [133] calculate the camera trajectory by using epipolar geometry to calculate the mapping between virtual points and stable positions in order to provide constraints for outputting stable video sequences. This technology also reduces the pressure on the long characteristic orbit. However, it requires a medium feature track length (usually over 20 frames).

Liu et al. [9] present bundled camera path model which maintains multiple, spatially-variant camera paths and directly uses the mesh vertices as the motion model itself. The method is designed to handle parallax and rolling shutter effects by local transformations and a shape-preserving constraint (see Fig. 10). However, the stabilized results occasionally contain visible blur artifacts. Specifically, let $\{p_i, \hat{p}_{i+1}\}$ be matched feature pair from frame i to frame $i + 1$. The feature p_i can be represented by a 2D bilinear interpolation of four vertices $V_p = [v_p^1, v_p^2, v_p^3, v_p^4]$ of closed mesh unit: $p = V_p w_p$, where w_p are interpolation weights. The authors expect that the corresponding feature $\hat{p}_{i+1} = \hat{V}_p w_p$ and the transformed vertices $\hat{V}_p = [\hat{v}_p^1, \hat{v}_p^2, \hat{v}_p^3, \hat{v}_p^4]$. Therefore the data term E_d is defined in Eq. (1). The shape preservation term E_s is used as [111].

$$\begin{aligned} E_d &= \sum_p \left\| \hat{V}_p w_p - \hat{p}_{i+1} \right\|^2 \\ E_s &= \sum_v \left\| \hat{v} - \hat{v}_1 - s R_{90} (\hat{v}_0 - \hat{v}_1) \right\|^2 \end{aligned} \quad (2)$$

where $s = \frac{\|\hat{v} - \hat{v}_1\|}{\|\hat{v}_0 - \hat{v}_1\|}$ and $R_{90} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. The final energy term E is formed by the linear combination of both terms: $E = E_d + \alpha E_s$, α is a regularization parameter. The local homography $T_j(i)$ of the grid cell j can be estimated by solving a linear equation $\hat{V}_{\hat{p}} = T_j(i) V_p$, which relates the four vertices before and after the warping. Zhao et al. [138] propose adaptively meshed video stabilization. According to literature [138], feature trajectories are extracted to generate a triangle mesh, and triangular mesh calculates the transformation between stabilized and shaky frames to stabilize the shaky video. Two adaptive weighting mechanisms are proposed to improve its spatial and temporal adaptability. The method can handle strong parallax scenes well. Wang et al. [139] propose a motion-direction-aware frame selection method to select stitching frames from video. It uses mesh-based frame warping to perform frame stitching and content stabilization. However, the method may fail when the matched feature lacks in frame regions and exists non-rigid object with inconsistent appearance. Wu et al. [140] present a novel method for video stabilization and rolling shutter removal. Based on literature [9], a neighbour-motion-aware local motion model which estimates a set of spatially variant inter-frame motion and adaptive weights are integrated into a motion-aware warping step to stabilize the results and remove rolling-shutter distortion.

(3) Global optimization method

Zhang et al. [136] implement motion compensation and image warp as a global optimization process rather than as two separate steps. Similar invariant constraints and spatially varying similar deformations are embedded into positional encodings in mesh vertices, which enables more flexible and efficient stabilization performance. Ma et al. [141] conclude that the combination way proposed by Zhang et al. [136] is not optimal. They propose a different way to combine trajectory smoothing and warping, trajectory smoothing plays a significant role and frame warping plays an auxiliary role to achieve more aggressive stabilization than Zhang et al. [136]. However, the method still cannot run online.

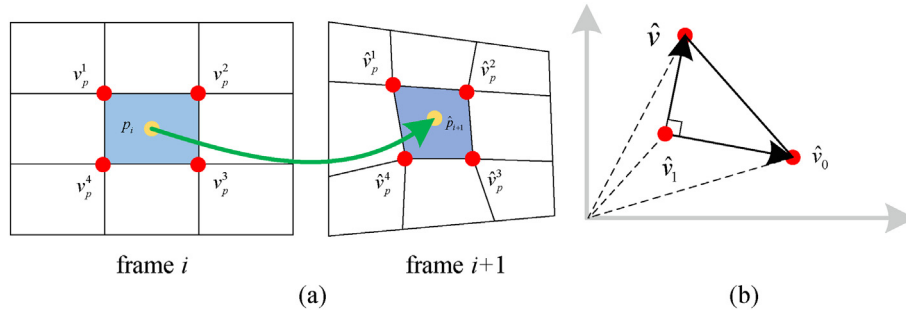


Fig. 10. (a) Keypoints-based mesh, adapted from Liu et al. [9] (b) The triangle of neighboring triangle $\hat{v}$, $\hat{v}_0$, $\hat{v}_1$ to follow a similarity transformation.

2.3.3. Image warp

For the 2.5D stabilization, it is necessary to perform image warp on the smooth motion trajectory to output a stable video. In the classical 2.5D stabilization proposed by Liu et al. [7], after the characteristic trajectory is multiplied by the coefficient matrix of the original video sequence to produce a smooth and stable trajectory, the final stable video is obtained by using the transformed image warp method based on content-preserving warps.

Wang et al. [134] adopt a video sequence-based rendering method to warp frames based on optimized feature locations to obtain stable videos. This method does not cause obvious visual artifacts. Zhang et al. [136] consider motion compensation and image warp as an overall global optimization process.

3. Deep learning-based digital video stabilization

With the developments of deep learning, more and more methods based on convolutional neural networks (CNNs) have been proposed for digital video stabilization in recent years. Instead of explicitly computing a camera path, these methods model a supervised learning approach. The CNN-based methods are more robust than traditional stabilization methods since they are adept at extracting high-dimensional abstract features, instead of relying on artificial features extracting and matching. Traditional DVS methods struggle with feature detection under certain conditions. The deep DVS methods are capable of extracting trajectory features.

In deep learning methods, according to whether depth information is considered, digital video stabilization is divided into *two-dimensional (2D)* scenes, *three-dimensional (3D)* scenes and *hybrid (2.5D)* scenes. The taxonomy of learning-based digital video stabilization is as follows in Fig. 12.

3.1. 2D scenes

In this section, we describe some methods based on deep learning for video stabilization. As is shown in Fig. 11, the aim is to find a network that can predict frames from the unstable version as sim-

ilar to the stable version as possible. The motion in the plane is considered in two-dimensional scenes, and the parallax problem is not solved. The proposed methods are mainly divided into trajectory optimization, frame interpolation and view synthesis.

3.1.1. Trajectory optimization

1) **StabNet [11]:** Wang et al. firstly present a convolutional neural network (CNN) called StabNet for video stabilization. The StabNet is based on supervised learning and it is a two-branch siamese network with sharing network parameters. A two-stage network is composed of a backbone encoder and a multi-grid regressor in each branch, which predicts the frame warping to produce the stabilized videos. The encoder is an adapted ResNet-50 [142] backbone model to extract high-dimension features and a multi-grid regressor is used to predict the stabilized transformation from extracted feature maps. Fig. 13 shows the network architecture.

The inputs of StabNet are an incoming shaky frame and six historical stabilized frames. The loss function includes three terms, namely stability, shape-preserving and temporal smoothness and is based on the adjacent frame I_i and I_{i+1} defined as:

$$L = \sum_{i \in \{t, t+1\}} L_{\text{stab}}(F_i, I_i) + L_{\text{shape}}(F_i, G_i) + L_{\text{temp}}(F_t, F_{t+1}, I_t, I_{t+1}) \quad (3)$$

where L_{stab} , L_{shape} and L_{temp} are the stability loss, shape-preserving and temporal loss. The model output must be a set of warping F . Inspired by bundled camera path [9], G_i is mesh vertex positions of the stabilized frame.

Another contribution is the DeepStab datasets. DeepStab consists of steady and unsteady video pairs. Details will be introduced in Section 5.2. The advantages of Stabnet are performing without using future frames in real-time and experiments show that fps is 35.5 on NVIDIA GTX 1080Ti graphic card. At the same time, It can stabilize low-quality videos. However, the disadvantage is that the proposed backbone ResNet-50 encoder does not consider the spatial changes, and the methods cannot handle the parallax problems.

2) **STNs [12]:** Xu et al. train GANs to produce the steady frame as guidance and use spatial transformation parameters to extract

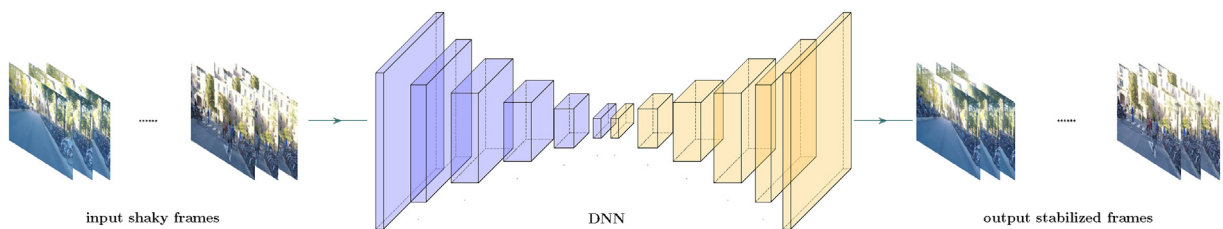


Fig. 11. Learning-based methods.

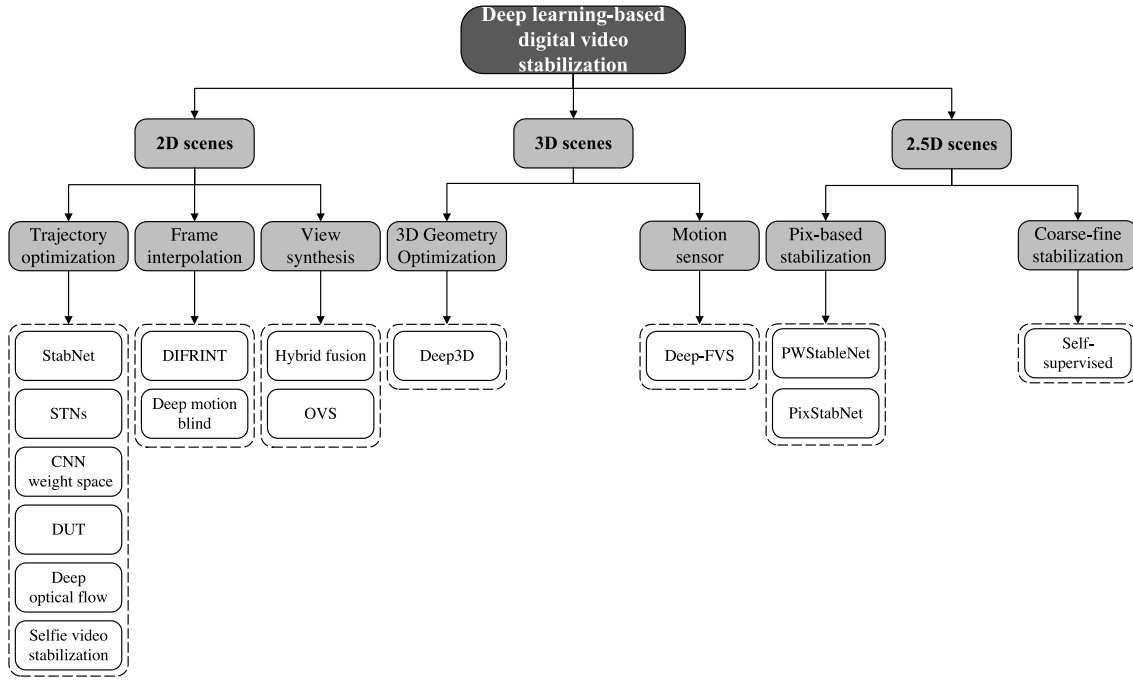


Fig. 12. A taxonomy of deep learning-based digital video stabilization.

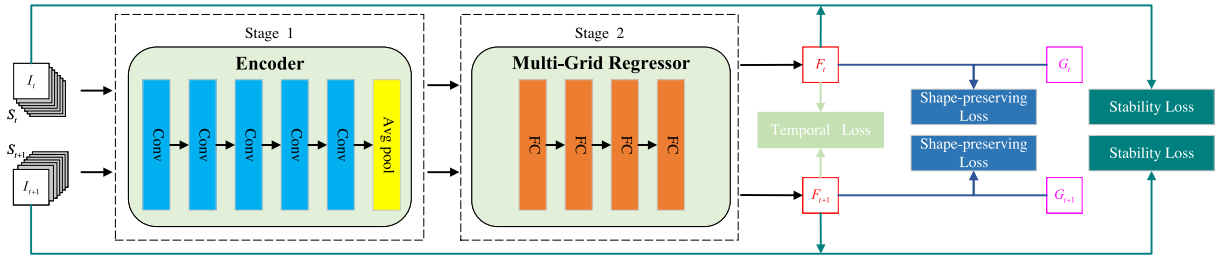


Fig. 13. Network architecture of the StabNet.

the affine transform for stabilizing the videos. A generator-discriminator architecture (GANs) is proposed to stabilize the video. Specifically, an encoder-decoder generator with spatial transformer networks (STNs) is designed to be embedded in the different layers to predict the steady frames for unsteady frames. Simultaneously, a discriminator network is defined to determine whether the generated frames are stable or shaky. The generated pixel-wise frame serves as the cropping free conditional input and the warping parameters are estimated by the adversarial network to generate stabilized frames. The stabilization loss is computed as:

$$L_{stab}(I_i, \hat{I}_i) = \lambda_1 \|Vgg^{19}(I_i) - Vgg^{19}(\hat{I}_i)\|_1 + \lambda_2 \|I_i - \hat{I}_i\|_1 \quad (4)$$

where I_i and $\hat{I}_i$ are the i th unsteady and stabilized frame respectively, Vgg^{19} is a pre-trained VGG 19-net. λ_1 and λ_2 are weighting parameters. As discriminator network, they use two 8 layers fully convolutional networks (FCNs) D_1, D_2 . Both network loss functions can be shown as:

$$\begin{cases} L_{D_1} = \|D_1(G_i, \bar{I}_i)\|_2^2 + \|1 - D_1(G_i, \hat{I}_i)\|_2^2 \\ L_{D_2} = \|D_2(A_i, \bar{I}_i)\|_2^2 + \|1 - D_2(A_i, \hat{I}_i)\|_2^2 \end{cases} \quad (5)$$

where G_i is a collection of the last five ground truth frames, $\bar{I}_i$ is the stable frames and A_i contains the last five frames generated by the generator. Finally, the generator's loss is:

$$L_G = L_{stab} + D_1(G_i, \hat{I}_i) + D_2(A_i, \hat{I}_i) \quad (6)$$

Xu et al. propose an adversarial network. The method is suitable for low-quality videos especially when the feature tracking is unreliable and the run time is 30.1 fps. However, the method ignores local transformations, temporal coherence is weak and the need to crop the frame boundaries and achieve less stability.

3) **CNN weight space [13]:** Yu et al. propose a novel video stabilization by optimization in CNN weight space. Unlike traditional CNN methods, the CNN weight space method does not use large datasets or pre-trained model and simply use CNN as a robust global optimization measure. The network architecture is as follows. (See Fig. 14)

The video is pre-stabilized by using basic 2D affine transformation firstly. Then, the optical flow is adopted between the adjacent frame in the pre-stabilized video and Yu et al. formulate the video stabilization as the minimization of the distance between corresponding pixels. Furthermore, A CNN is used as the optimizer to solve for 2D affine transformations and warp fields. Finally, videos are warped and cropped to rectangles to produce stabilized results. The final objective function is written as:

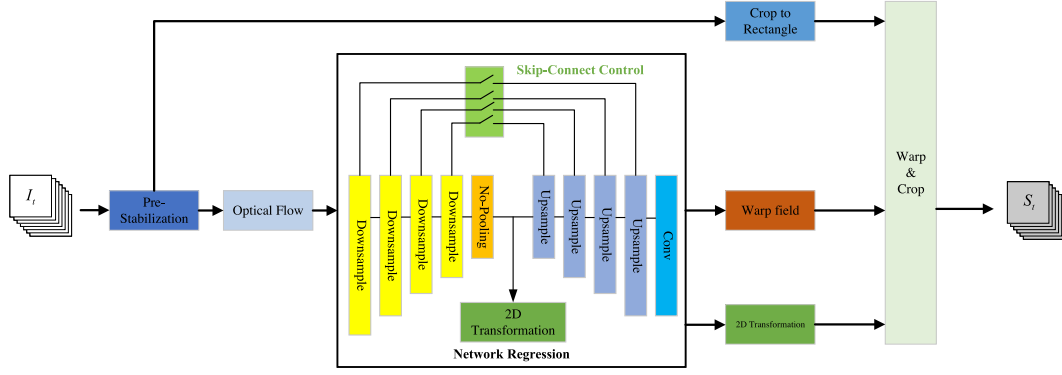


Fig. 14. Network architecture of the CNN weight space.

$$L = \min_{\theta} E_o(\mathbf{G}(\theta)) + \lambda \|E_p \cdot E_r(\mathbf{G}(\theta))\|_1 \quad (7)$$

where E_p represents the original optical flow, E_r is the regularization, E_o is the Euclidean distance between the warped pixel positions and λ is a hyperparameter controlling the amount of regularization. $\mathbf{G}(\theta)$ is the neural network, and θ represents the parameters of the network. The goal is to get the parameters θ .

The method can be generalized to test videos without optimization. However, the limitation is offline method that requires 30 min to stabilize 300-frame video on a GTX1080Ti graphics card.

4) **DUT** [16]: Xu et al. propose a deep unsupervised trajectory-based stabilization framework (DUT), which contains three modules: keypoint detection, motion propagation and trajectory smoothing. In the keypoint detection, the RFNet [143] and PWCNet [40] are utilized for motion estimation. In motion propagation, the motion vectors are propagated from sparse key points to dense grid vertices by DNNs. In smoothing trajectory, the estimated trajectories are smoothed at all grid vertices by predicting kernel weights. The network architecture of motion propagation and trajectory smoothing is followed in Fig. 15, and the objective function is shown in Eq. (8).

$$\begin{cases} L_{vm} = \lambda_m \sum_{i=1}^{E-1} \sum_{k=1}^M \sum_{j \in \Omega_{ik}} \|n_{ik} - m_{ij}\|_1 + \lambda_v \sum_{i=1}^{E-1} \sum_{j=1}^L \|p_{ij} + m_{ij} - H_{ij}(p_{ij})\|_2^2 \\ L_{sp} = \sum_{i=1}^{E-1} \sum_{m=1}^{(M-1)(N-1)} \|\hat{v}_{im}^3 - (\hat{v}_{im}^2 + R_{90}(\hat{v}_{im}^1 - \hat{v}_{im}^2))\|_2^2 \end{cases} \quad (8)$$

where L_{vm} is motion vectors estimation loss and L_{sp} presents shape preserving loss inspired by bundled camera path [9]. In the L_{vm} , n_{ik} is target motion vector. m_{ij} is the motion vector of keypoint p_{ij} , H_{ij} is homography transformation, L is the number of detected keypoints, E is the number of frame and R_{90} is same as in Eq. (2). In the L_{sp} , $\hat{v}_{im}^i = v_{im}^i + n_{im}^i$, $i = 1, 2, 3, 4$, v_{im}^i denote four vertices, n_{im}^i are motion vectors. The final motion propagation loss L is form by both terms: $L_{MP} = L_{vm} + \lambda_s L_{sp}$ and λ_s is loss weight.

The objective function of trajectory smoothing is shown on Eq. (9).

$$\begin{cases} L_{ts} = \sum_{i=1}^{E-1} \left(\|\hat{T}_{ik} - T_{ik}\|_2^2 + \lambda \sum_{j \in \Omega_i} w_{ij} \|\hat{T}_{ik} - \hat{T}_{jk}\|_2^2 \right) \\ L_{cp} = \sum_{i=1}^{E-1} \sum_{j=1}^L \|\text{Bili}(p_{ij}) - H_{ij}(p_{ij})\|^2 \end{cases} \quad (9)$$

where L_{ts} is the temporal smoothness loss, in the L_{ts} , $\hat{T}_{ik}$ denotes the smoothed trajectory at v_{ik} , T_{ik} is the corresponding grid vertices formulates trajectories. w_{ij} is the weight controlling trajectory smoothness, λ balances the first term and second term. Ω_i is the temporal neighborhood of frame. L_{cp} is the content preserving loss. In the L_{cp} , H_{ij} and p_{ij} is the same as in Eq. (8). $\text{Bili}(\cdot)$ is a bilinear interpolation. The final objective function is: $L_{TS} = L_{ts} + \lambda_s L_{sp} + \lambda_c L_{cp}$, the L_{sp} is the same as in Eq. (8), and λ_s, λ_c is the weight loss.

DUT is proposed for the first time in an unsupervised manner and generates stabilized video with less distortion. Nevertheless, if one of the planes is texture-less, DUT may result in distortion.

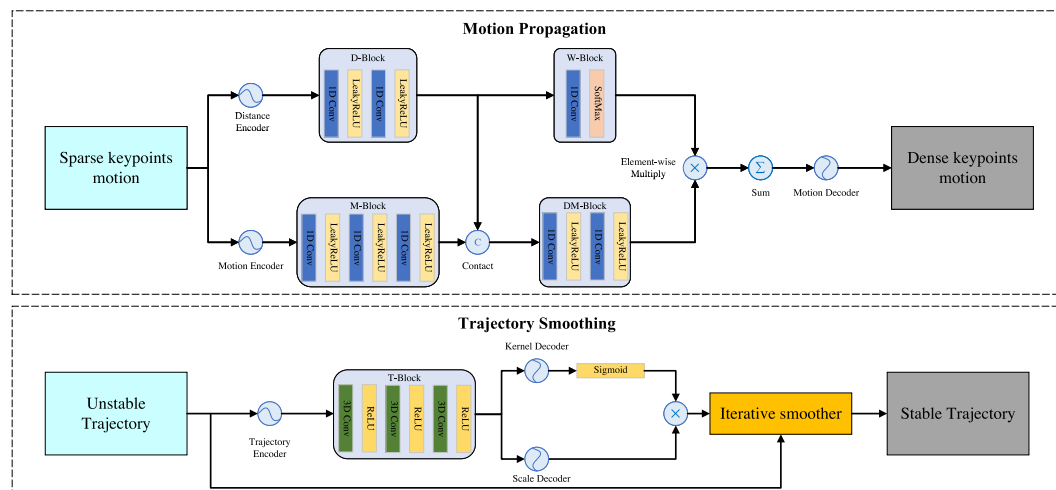


Fig. 15. Network architecture of the DUT.

5) **Deep optical flow** [14]: Yu et al. propose a learning video stabilization using optical flow. The contribution of the algorithm is a deep neural network that takes the optical flow as the input and directly outputs the warp fields. The method is divided into three stages. In the first stage, SURF features [80] are computed and stabilized with translation and rotation. Then, the optical flow between consecutive frames is computed with FlowNet2 [144]. A mask is generated to indicate the valid regions and PCA flow is proposed to inpainted the invalid regions. In the second stage, a stabilized network is proposed. The input is the inpainted optical flow field, and the network generates a per-pixel warp field to compensate for the frame motion. In the third stage, PCA flow is fitted to the raw warp field and warped video is cropped to a rectangle as output. The loss function is summarized into two phases.

$$L_1 = L_m + 10 * L_f \quad L_2 = L_m \quad (10)$$

where L_1 is the loss function in the first training phase and L_2 is the loss function in the second training phase. L_m is the motion loss and L_f represents frequency domain loss.

The Deep optical flow is more robust and is compared to the optimization-based method with a $\sim 3\times$ speed improvement.

6) **Deep Selfie video stabilization** [20]: Yu et al. propose a novel real-time selfie video stabilization. The pipeline is composed of three parts: motion detection, stabilization and warping. Specifically, A segmentation network FCN8 is trained to infer the foreground and remove feature points in the foreground. Then, the background motion is tracked by KLT and the foreground motion is tracked by 3D face vertices. Furthermore, a two-branch 1D linear CNN is designed to infer the displacement of the moving least square (MLS) warp notes. Finally, A grid is used to approximate the dense MLS warping to run at a real-time rate and produce the stabilized frame. The loss function of the stabilized network is as follows:

$$L = (1 - \lambda)L_b + \lambda L_f \quad (11)$$

where L_b denotes the background loss, L_f denotes the foreground loss and $\lambda \in (0, 1)$ controls the stabilization focus.

The method infers the moving least square warp from the 2D feature points and can stabilize the video of resolution 832×448 at 26 fps on an NVIDIA 2080Ti graphics card. But if very few feature points are detected in the background, the method may fail.

Another contribution is selfie video datasets with pre-frame labelled foreground masks. The selfie video dataset contains 1005 videos. Details will be discussed in Section 5.2.

3.1.2. Frame interpolation

1) **DIFRINT** [17]: Jinsoo et al. propose the Deep Iterative Frame Interpolation (DIFRINT) in an unsupervised manner. The key idea of their approach is to iteratively interpolate a middle frame between two frames and produce a stabilized video without cropping, which leads to visual stability. A deep frame interpolation architecture is as shown. (See Fig. 16)

Within DIFRINT framework, the adjacent frame I_{i-1} and I_{i+1} is warped towards the I_s and aims to reconstruct I_s . From two frames, I_w^- and I_w^+ are computed from a bidirectional optical flow (PWC-Net). These frames are fed to a U-Net [145] architecture to produce the intermediate frame I_{int} , which is used as a reference with I_s . Furthermore, avoiding blur effects, I_i is fed along with I_{int} to ResNet [146] to generate $\hat{I}_i$, which is also used as a reference with I_s . They use the pixel-wise color-based loss function. The loss L_{int} and L_{out} are as follow.

$$\begin{cases} L_{int} = \|I_s - I_{int}\|_1 + \|\phi(I_s) - \phi(I_{int})\|_2^2 \\ L_{out} = \|I_s - \hat{I}_i\|_1 + \|\phi(I_s) - \phi(\hat{I}_i)\|_2^2 \end{cases} \quad (12)$$

where L_{int} is the U-Net loss and L_{out} is the whole network loss separately. $\phi(\cdot)$ is the response from the relu4_3 layer of VGG-19 and I_s is the pseudo-ground truth frame.

The proposed framework implicitly takes advantage of the spatial coherence between adjacent frames and the content in each frame. It utilizes frame interpolation to generate stabilized frames, thereby reducing inter-frame jitter. Once applied iteratively, the stabilization becomes more robust. The approach enables full-frame (without cropping) video stabilization in near real-time (14.3 fps) using two NVIDIA TitanX(Maxwell). However, applying frame interpolation repeatedly leads to severe distortion and blur artifacts in challenging cases.

2) **Deep motion blind** [18]: Ali et al. propose deep motion blind video stabilization. Considering the existence of perspective inconsistency in Deepstab, the existence of substantial cropping and temporal distortions in warping methods, an unsupervised dataset generation technique is proposed. The data generation technique is used to produce equi-perspective stabilized videos. It contains an iterative frame interpolation and refinement network. The SOTA interpolation CAIN [147] is introduced to compensate for high-frequency shaky. Meanwhile, a refinement network is introduced to remove the artifacts, which finally generates video stabilization datasets. Furthermore, a training strategy is proposed. It contains three stages: stable frame generation, quality enhancement and strengthening. The proposed method can maintain less distortion and less cropping. It achieves real-time run speed.

3.1.3. View synthesis

1) **Hybrid fusion** [15]: Liu et al. propose a hybrid fusion mechanism that it consists of the feature extractor, warping layer, fusion function frame generator and the final weighted sum step to generate the output video. Specifically, the author uses a CNN to encode each frame and then leverages both feature-level and image-level fusion to alleviate the sensitivity to flow inaccuracy and learns to predict spatially varying blending weights to remove blurry frames and transfer high-frequency details residual to the re-rendered stabilized frames. Furthermore, A path adjustment method is proposed to balance the smoothing camera motion and maximize frame coverage. The loss function includes the L1 and VGG perceptual losses.

$$L = \|I_i - \hat{I}_i\|_1 + \sum_l \lambda_l \|\psi_l(I_i) - \psi_l(\hat{I}_i)\|_1 \quad (13)$$

where I_i is the shaky frame, $\hat{I}_i$ denotes the ground-truth target frame and ψ_l is an intermediate feature from a VGG-19.

The proposed method generates full-frame stabilized videos with fewer visual artifacts and can be easily integrated with existing flow smoothing methods to further stabilize. However, the limitation is useless for rolling shutter and the results may fail in some cases, such as lighting variations, occlusion and foreground/background motion. The method is designed for offline applications and runs slowly (10 s per frame).

2) **OVS** [21]: Xu et al. propose a new out-of-boundary view synthesis (OVS). It can significantly reduce cropping and resizing. OVS consists of coarse and fine stages. In the first coarse alignment stage, the successive frames use a grid-based motion estimate to align. Namely, the motion estimation module in DUT is taken. In the second fine alignment stage, the optical flow is calculated and then use affinity propagation to predict the flow in the out-of-boundary views. Finally, the successive frames are warped. This procedure is iteratively implemented to gradually align distant neighbouring frames to extrapolate the area. The fine alignment stage is expressed as follows. (See Fig. 17)

The first image on the left shows the coarse result. Two images in the middle show that the estimation of the optical flow Op_{i-i+1}

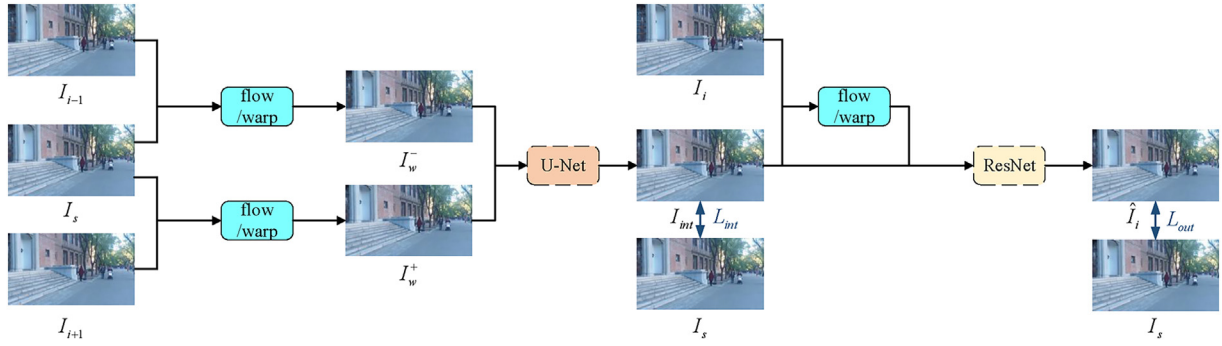


Fig. 16. Interpolation module on training.

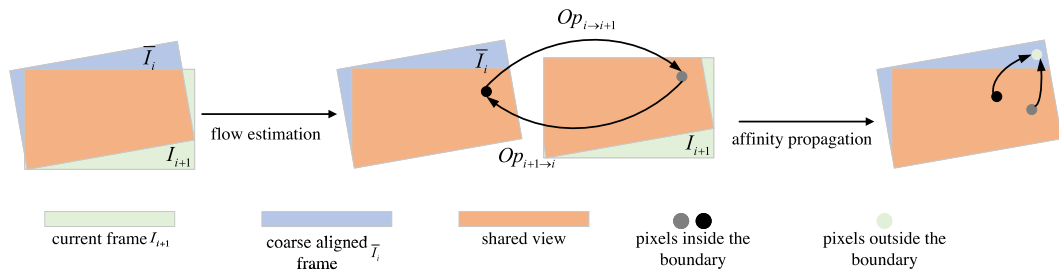


Fig. 17. The fine alignment stage of OVS.



Fig. 18. Example of NUS datasets.

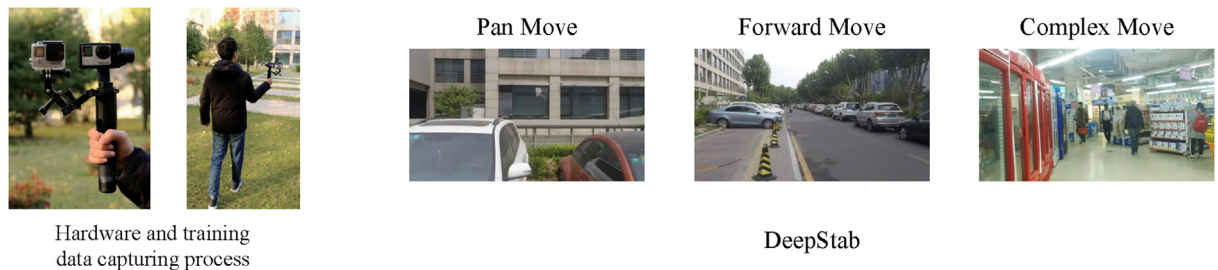


Fig. 19. DeepStab datasets and capturing process.



Fig. 20. Part of the selfie video examples [20].



Fig. 21. Part examples of the video + sensors dataset [23].

from current frame I_{i+1} to the aligned frame $\bar{I}_i$ and reverse flow $\bar{O}p_{i+1-i}$ from $\bar{I}_i$ to I_{i+1} . Finally, the out-of-boundary pixels can be extrapolated by using affinity propagation. The loss function contains two parts. The loss function in the coarse stages is the same as the DUT [16]. The loss function in the fine stages is $L = L_I + 2 \times L_G + 2 \times L_M$, where L_I is the prediction loss, L_G is the ground truth loss and L_M is the regularization loss.

Despite the adopt supervised training for OVS, it does not require paired data. The proposed method can help pixel-based and grid-based stabilizers to achieve near full frames with less distortion and high stability. However, some visible seams may exist in some cases.

3.2. 3D scenes

3.2.1. 3D Geometry optimization

Deep3D [22]: Inspired by the self-supervised framework of Zhou et al. [148], Lee et al. firstly propose a novel 3D-based deep learning method for video stabilization (Deep3D) without needing training data, which can handle parallax problems and severe camera shakes to achieve better stability and low distortion. The pipeline mainly consists of 3D geometry optimization and frame rectification. Firstly, 3D geometry optimization contains PoseNet and DepthNet, which estimates the 3D camera trajectory and dense scene depth. Secondly, the frame rectification utilizes the estimated trajectory and dense scene depth to perform view synthesis on a smoothed trajectory. Finally, resulted video is then wrapped and cropped to generate the stabilized video. The loss terms consist of three terms: photometric loss L_p , geometric loss L_G and optical flow loss L_F in the 3D geometry optimization framework.

$$\begin{cases} L_p(i, s) &= \frac{1}{|V|} \sum_{x_i \in V} (1 - \gamma) \|I_{s-i}(x_i) - I_i(x_i)\|_1 + \gamma \frac{1 - SSIM_{I_{s-i}I_i}(x_i)}{2} \\ L_G(i, s) &= \frac{1}{|V|} \sum_{x_i \in V} \frac{\|D_{s-i}(x_i) - D_i(x_i)\|_1}{D_{s-i}(x_i) + D_i(x_i)} \\ L_F(i, s) &= \frac{1}{|V|} \sum_{x_i \in V} \|F_{s-i}(x_i) - \hat{F}_{s-i}(x_i)\|_1 \end{cases} \quad (14)$$

where L_p measures the appearance difference between synthesized I_{s-i} and frame I_i and structural similarity loss with the weight γ . V is the valid points successfully and $|V|$ represents the point numbers. L_G enforces the scale consistency of depth and pose estimation, depth maps D_{s-i} are employed for 3D scene reconstruction and

camera motion estimation and D_i is scene depth. Pre-trained PWC-Net [40] is used to compute the difference between the estimated F_{s-i} and precomputing flow $\hat{F}_{s-i}$ by L_F .

Deep3D can run in real-time (34.5 fps) on NVIDIA RTX2080Ti. But the Deep3D cannot handle the intra-frame effects (eg. rolling shutter and blur scene) and the crop area is too large.

3.2.2. Motion sensor

Deep-FVS [23]: Shi et al. propose an efficient deep Fused video stabilization. It is the first deep learning method that adopts both gyroscope data and optical flow for video stabilization. Given input videos, Deep-FVS deals with the gyroscope and OIS reading to extract real camera poses and the raw optical flow. Then, optical flow is encoded to a latent representation via 2D convolutions and concatenated with real and previous camera poses. Furthermore, the joint motion representation is fed to an LSTM cell and FC layers to predict the new virtual camera poses. Finally, the input frame based on OIS movement and the predicted virtual camera poses is wrapped to generate the stabilized frame by a grid-based method like [149]. The loss function consists of smoothness, protrusion, distortion and optical flow loss. In addition, A multi-stage is adopted to train the network.

Another contribution is the Video + Sensor datasets, which contain 50 videos with sensor logs. Details will be introduced in Section 5.2. Unlike image-based methods, Deep-FVS requires sensor data and can handle parallax problems. However, given 1920×1080 resolution videos, the fps is 3.5 on Intel Xeno CPU and Tesla T4 GPU.

3.3. 2.5D scenes

3.3.1. Pix-based stabilization

1) **PWStableNet [24]:** Zhao et al. propose a novel video stabilization network called PWStableNet, a supervised method. The method is based on a siamese network and it has two branches to share the parameters. The method contains three stages and each stage is mainly an encoder-decoder framework. Multi-stage cascade encoder-decoder architecture is designed to estimate horizontal and vertical warping maps. Then, inter-stage connections are also introduced to enable the latter stages to learn residual about former stages. This architecture can produce more precise warping maps. The loss function is defined as follows. n -th stage is taken as an example.

$$L_{stab}^n = \sum_{i \in \{t-1, t\}} L_{content}(\hat{I}_i^n, \tilde{I}_i) + L_{shape}(T_i^n, \tilde{I}_i, I_i) + L_{temporal}(\hat{I}_{t-1}^n, \hat{I}_t^n) \quad (15)$$

where $L_{content}$ is content loss, stabilized frame $\hat{I}_i^n$ is pushed to approximate its groundtruth $\tilde{I}_i$. L_{shape} is the shape loss to preserve the shape of warped frame. T_i^n is the warping maps and I_i is unstable frame. $L_{temporal}$ is temporal loss and it aims to require the smoothness of successive stabilized frame. Finally, the loss function of all stages is $L_{stab} = \sum_{n \in \{1, 2, 3\}} L_{stab}^n$.

The PWStableNet can handle parallax well, and has stronger robustness and higher processing speed (56.2 fps) on NVIDIA GTX 1080Ti. But, large quick motion or strong parallax may lead to performance degradation and the bottleneck of PWStableNet lies in the capacity of the training dataset. Furthermore, it may incur a delay of several frames, resulting in a fixed delay.

2) **PixStabNet [25]**: For several frames delay in PWStableNet, Chen et al. propose a fast multi-scale online video stabilization method based on pixel warping. Firstly, a multi-scale CNN is proposed to predict transformations. Then, they introduce the encoder-decoder architecture to estimate pix-based warping maps to stabilize video. The loss function is composed of three terms: stability loss L_{stab} , distortion-reducing loss L_{dr} and temporal loss L_{temp} .

$$L^s = \sum_{k \in \{t-1, t\}} \alpha L_{stab} + \beta L_{dr} + \gamma L_{temp} \quad (16)$$

where α, β and γ is the weight. Due to the introduction of scale information s , the total loss function is expressed as $L_{total} = \sum_{s \in \{0, 1, 2\}} 2^s L^s$.

Furthermore, a two-stage training scheme is designed to contribute to more robust stabilization results.

PixStabNet can deal with parallax problems and has geometric and temporal consistency. It can run at 54.6 fps without future frames in NVIDIA RTX 2080Ti. Nevertheless, It may cause large cropping due to stabilization.

3.3.2. Coarse-fine stabilization

Self-supervised [26]: Jinsoo et al. propose a self-supervised real-time video stabilization. The cascade module is introduced, which consists of three modules, coarse, fine stabilizer and margin inpainter. The Coarse-Net is designed to estimate the rigid transformation for stabilization globally. Then, the Fine-Net is designed to estimate the spatially smoothed flow to remove the remaining shaky. Finally, the Margin-Net is introduced to fill in the frame boundaries to reduce the cropping. Furthermore, a self-supervised learning scheme is proposed to train all three networks appropriately.

$$\begin{cases} L_C &= \alpha L_t + \beta L_p + \gamma L_n \\ L_F &= \alpha L_f + \gamma L_n \\ L_M &= L_g + L_l \end{cases} \quad (17)$$

where L_C is the Coarse-Net loss, L_t measures the L1 loss between randomly and the estimated transformation by the Coarse-Net, L_p measures the L1 loss between ground truth and estimated parameter vectors, L_n is the L1 loss between the input frame and estimated frame, which to learn roughly aligning successive frames. L_f is the Fine-Net loss, L_f measures the L1 loss between randomly and estimated transformation by the Fine-Net, and L_n measures the L1 loss between the input frame and warped frame by using Fine-Net's predicted smoothed flow. L_M is the Margin-Net loss, L_g measures L1 loss between the transformed frame and inpainted image, L_l measures the L1 loss focus only on the inpainted region.

The method outperforms various videos, such as parallax, rotation. It runs in real-time and achieves low levels of cropping, low visual distortion and high stability.

In summary, deep learning-based methods are very promising and have quickly become a trend. Despite there are not so many deep learning-based methods now, existing learning-based methods show powerful non-linear expressions and high-quality results. Thus, it is necessary and possible to develop methods that outperform traditional methods. However, learning-based methods suffer from the variety of application scenarios. For scenes lacking parallax, the network architecture should be designed without estimating scene depth. whereas for selfie scenes, the loss function should contain foreground and background loss, which is different from the loss function in general scenes.

4. Digital video stabilization quality assessment

In video stabilization, the visual discomfort caused by camera irregular motion and the artifacts caused by the stabilization process will lead to video quality degradation. Video stabilization quality assessment is mainly to evaluate video sequence quality, which is the main indicator to measure advanced and cutting-edge of video stabilization. Prior work [150–154] have been reported about study on performance assessment of video stabilization. However, there is currently no unified benchmark framework for vs quality assessment.

This section reviews the development of main video stabilization quality assessment. Similar to the video quality assessment method, the video stabilization quality evaluation is also divided into two categories: subjective and objective assessment. Two main assessment methods have been used to evaluate video stabilization results.

4.1. Subjective quality assessment

Subjective assessment involves people's subjective assessment based on the human visual system. It assesses video by visual inspection or user studies.

Mean Opinion Score (MOS) [155] is currently a common method for subjective evaluation. It is firstly proposed by Suan et al. [156] and is established on the basis of statistics in medicine. In order to ensure that the subjective evaluation of video has statistical significance, the number of observers should be enough, and its subjective evaluation results can truly reflect the video stability. However, the mathematical model cannot express the MOS evaluation method, and it is time-consuming. Therefore, it is not suitable for the stability evaluation of large datasets. The Mean Opinion Score difference (Differential Mean Opinion Score, DMOS), DMOS is established based on the MOS Score. It describes the difference between the human eye's evaluation score of distortion-free and distorted images. The formula is as follows:

$$\begin{cases} d_{ij} &= MOS_{original} - MOS_{distorted} \\ d'_{ij} &= \frac{d_{ij} - \min(d_{ij})}{\max(d_{ij}) - \min(d_{ij})} \end{cases} \quad (18)$$

where d_{ij} represents the difference between the evaluation scores of the human eye for the undistorted image and the distorted image, and finally d'_{ij} takes the average value to obtain the DMOS value. Likewise, the DMOS method is unsuitable for assessing large datasets.

Furthermore, In the literature [5], they explain the user study based on MOS to measure the subject preference based on the paired comparison between different methods. Specifically, observers are shown several videos and are asked to select the best one according to some predefined metrics. However, a user study is

also time-consuming and difficult to construct based on a mathematical model. At present, it is necessary for subjective assessment, especially since there is no unified objective benchmark framework for evaluation.

4.2. Objective quality assessment

The objective assessment method uses metrics that can reflect the video stabilization quality to establish a mathematical model and then assess the video stabilization. According to whether there is a reference object, it is subdivided into full-reference and no-reference methods. The full-reference method compares the stabilized and stable videos recognized as the reference metrics. The no-reference method does not require stable videos and relies on statistical models to evaluate the motion changes during the stabilized video.

4.2.1. Full-reference methods

For full-reference methods, the stable video is often used as a reference metric, and the shaky videos are synthesized with stable videos by adding jitter. The difference between the stable and stabilized video is used to measure the quality of the video stabilization. Common full-reference assessment methods are Peak Signal-to-Noise Ratio (PSNR) [157], Mean Squared Error (MSE) [158], and Structural Similarity (SSIM) [159]. They are widely adopted for their applicability, computability and are suitable for integration into optimization algorithms.

Offiah et al. [160] propose a full-reference quality assessment method for the stabilization of medical endoscopic videos. Tanakian et al. [152] use the mean square error (MSE) between the stabilized motion path and the ground truth motion path as a distance metric to evaluate the video stabilization quality. But, two assessments may be inconsistent with the results of the human visual system. Qu et al. [161] propose a full-reference dataset that consists of shaky and stable videos, and then use the structural similarity index (SSIM) to evaluate the processing effect of the video stabilization. But, the methods cannot reflect more stabilization characteristics. Zhang et al. [162] propose a video stabilization quality assessment algorithm based on the Riemann metric. The method uses a Riemann metric defined on a spatially transformed manifold to measure and assess the motion difference between the stabilized video and the ground truth. The accuracy of the method is better than others, but the computational complexity is also more remarkable than the others. Liu et al. [15] use PSNR, SSIM and LPIPS [163] to evaluate the quality of synthesized frames, proving the proposed methods' advantages. Wang et al. [164] propose a full-reference assessment method which combines the Distance, PSNR and SSIM for evaluation metrics. Ito et al. [165] propose an evaluation framework and introduce a comprehensive dataset for video stabilization. The proposed evaluation framework consists of MSE, SSIM, Distance between features, resolution loss and motion parameters MSE_{PR} . However, the method does not explore all characteristics of stabilization.

Due to the lack of large paired datasets, the full-reference method is not widely adopted in video stabilization.

4.2.2. No-reference methods

No-reference method does not need paired videos. It relies on a statistical model to measure the motion changes during video stabilization.

Niskanen et al. [166] use a low-pass/high-pass filter to separate the shaky part of the stabilized video and then measure its stability through jitter attenuation. However, relative attenuation does not make a clear judgment on the motion stability of stable video. In order to assess the stable video quality, Liu et al. [9] propose cropping, distortion and stability objective metrics. Cropping

measures the remaining frame area after cropping off the undefined pixels due to motion compensation. Distortion measures the anisotropic scaling of the homography between the input and output frames. Stability measures the stability and smoothness of the stabilized video. The metrics are widely used in prior work. However, this method needs to set the separation threshold of high/low frequency, which usually changes with the motion types of different videos. Zheng et al. [167] propose a no-reference evaluation method based on the total curvature of the motion path. The homography matrix is constructed by detecting the feature points, and it is mapped in Lie group space to form the motion path. Finally, discrete geodesic is used to calculate the total curvature of the motion path and assess the video's stability. On the basis of Zheng's work, Zhang et al. [168] use a constrained path to deal with spatially varying motion, calculate weighted curvature and evaluate the stability. Liu et al. [15] use accumulated optical flow metrics to measure the motion for video stabilization assessment.

The current quality assessment methods only deal with cropping, distortion and path factors, which can be used as loss function in learning-based optimization stage and also for assessment at the same time. However, these metrics cannot embody the perceptual aspects of video stabilization. The illumination conditions, the characteristics of the vision system and the accurate recognition of background area will affect the motion stability. In future work, more characteristics need to be explored and solved in order to further improve the accuracy of objective evaluation methods. Finally, a summary table is shown as follows. Table 3 describes the start-of-the-art approaches for video stabilization along with their description, advantages and limitations.

5. Performance comparisons

5.1. Public datasets

Datasets have played an important role in video stabilization. Various datasets have been created to motivate researchers to propose methods with better performances during the past years. Details of these most representative datasets used in video stabilization tasks are summarized in Table 4.

1. The **HUJ** dataset proposed by Goldstein et al. [133] consists of 42 video sequences, which are mainly driving, dynamic, zooming and walking scenes.
2. The **NUS** dataset consists of 144 video sequences, which are classified into seven categories: simple, quick rotation, zooming, parallax, driving, crowd, and running. (See Fig. 18)
3. The **MCL** dataset proposed by Koh et al. [135] consists of 162 videos, which are divided into seven categories: simple, rolling shutter, depth, crowd, driving, running and object scenes.
4. The **BIT** dataset proposed by Zhang et al. [162] consists of 45 videos, which includes walking, climbing, running, riding, driving, large parallax, crowd, near-range object and dark scenes.
5. The **DeepStab** dataset consists of 61 pairs of synchronized videos whose length is within 30 s at 30 fps. The datasets include indoor scenes with large parallax, and common outdoor scenes with buildings, vegetation, crowd, etc. Camera motions include forward movement, pan movement, spin movement and complex movements including combinations of the above at various speeds.
6. The **QMUL** dataset proposed by Ito et al. [165] consists of 421 videos, which involve simple, blurry, high motion, dark, textureless, parallax, discontinuous, depth, crowd and close object scenes.
7. The **Selfie** dataset proposed by Yu et al. [20] contains 1005 videos that develop on the basis of [19]. The Dlib is used to

Table 3

Summary of the most representative approaches for video stabilization.

Paper	Year	website	Description	Advantage	Limitations
Subspace [7]	2011	https://github.com/higerra/SubspaceStab	Subspace constraints on feature trajectory.	Reduce the requirement of long feature tracking and can handle parallax.	May fail in scenes lack of parallax.
L1Stabilizer [8]	2011	https://github.com/ishit/L1Stabilizer	Camera motion based on multiple constraints.	Views experience close to professional footage.	cropping discards information and unreliable in extreme scenes.
Bundle [9]	2013	https://github.com/SuTanTank/BundledCameraPathVideoStabilization	Mesh-based, space-time optimization.	Handle parallax without SfM and avoid rolling shutter distortion.	Failed to handle severe occlusions or distortion.
Meshflow [10]	2016	https://github.com/sudheerachary/Mesh-Flow-Video-Stabilization	Adopt Spatial smooth sparse motion field and work online.	Handle spatially-variant motion and also works online.	Cannot handle larger near-range foreground objects.
StabNet [11]	2018	https://github.com/cxjyxxme/deep-online-video-stabilization-deploy	Learn warping based on CNN.	Fast computation.	Fail to handle drastic motion and scenes with parallax.
STNs [12]	2018	https://github.com/mindazhao/Deep-Video-Stabilization-Using-Adversarial-Networks	Exact affine transform based on spatial transformer networks(STNs).	Suitable for low quality videos.	Ignore local transformations.
Selfie VS [19]	2018	-	Foreground and background separation, minimizing background smoothness.	Robust for selfie stabilization.	Fail in complex motion, motion blur and lighting condition.
CNN weight space [13]	2019	-	CNN as a robust global optimization measures.	Not require a large dataset.	offline.
DIFRINT [17]	2019	https://github.com/jjinsc37/DIFRINT	Iteratively interpolate a middle frame.	Full-frame(without cropping).	Distortion and artifacts in challenging cases.
Deep optical flow [14]	2020	https://github.com/posgraph/coupe.optical_flow_based_deep_video_stabilization	Optical flow is taken as the input and directly output the warp field.	Robust to optimization method with ~3x speed.	Fail in complex structures and quick object motion.
PW-StableNet [24]	2020	https://github.com/mindazhao/PWStableNet	Multi-stage cascade encoder-encoder is designed to estimate warping maps.	Handle parallax, has stronger robustness and higher processing speed.	Degradation in strong parallax scene.
DUT [16]	2020	https://github.com/Annbless/DUTCode	Deep unsupervised trajectory-based method.	Less distortion.	Degradation in texture-less planes.
Hybrid [15]	2021	https://github.com/alex04072000/FuSta	Based on hybrid fusion mechanism.	Full-frame and can be integrated with existing methods.	offline
Deep Selfie VS [20]	2021	https://github.com/jiy173/selfievideostabilization	Method based on 1D linear CNN.	Fast computation.	Fail in few feature detected scene.
DUT + OVS [21]	2021	https://github.com/Annbless/OVS_Stabilization	Coarse and fine stages to affinity propagation.	Full frame with less distortion and high stability.	Visible seams exist in some cases.
Deep3D [22]	2021	-	3D geometry optimization and frame rectification.	Handle parallax and severe camera shakes.	cannot handle rolling shutter and blur scenes.
PixStabNet [25]	2021	https://yu-ta-chen.github.io/PixStabNet	Multi-scale method based on pixel warping.	Fast computation.	Large cropping.
Deep-FVS [23]	2022	https://github.com/googleinterns/deep-stabilization	Based on both gyroscope data and optical flow.	Handle parallax problems.	Slow running speed.
Deep motion blind [18]	2022	https://github.com/MKashifAli/Motion_Blind_Video_Stabilization	Iterative frame interpolation and refinement network.	Less distortion and less cropping.	Fails in some challenging cases.
Self-Supervised [26]	2022	-	Based on Coarse-Net, Fine-Net and Margin-Net.	Fast computation regardless of resolution.	Artifacts in some cases(zooms, rotation).

Table 4
video stabilization datasets.

Datasets	Year	Download Link	Video Number	Resolution
HUJ [133]	2012	https://www.cs.huji.ac.il/w~raananf/projects/stab/videos/	42	1280*720, 640*360
NUS [9]	2013	http://liushuaicheng.org/SIGGRAPH2013/database.html	174	640*360
MCL [135]	2015	http://mcl.korea.ac.kr/stabilization/	162	1280*720, 640*360
BIT [162]	2018	https://zhangleibit.github.io/download/stabfr/dataset.html	45	-
DeepStab [11]	2019	https://cg.cs.tsinghua.edu.cn/people/~miao/stabnet/data.zip	61	1280*720
QMUL [165]	2019	https://github.com/mariito/DVS_	421	-
Selfie [20]	2021	https://mega.nz/folder/ftVmSTQS#iWV1Kunz0FGLA4b51A7Q0g	1005	1280*720
Video + Sensor [23]	2022	https://storage.googleapis.com/dataset_release/all.zip	50	1920*1080

detect faces in each frame and consecutive frames containing faces are counted. If the face can be detected for over 50 consecutive frames, the raw video will be cut into a new clip. It includes regular color videos and a ground truth foreground mask for each frame. (See Fig. 20)

8. The **Video + Sensor** dataset proposed by Shi et al.[23] which contains 50 videos with gyroscope and OIS sensor logs. The dataset is split into 16 videos for training and 34 videos for testing, and the test set is classified into 6 categories: general, rotation, parallax, driving, people and running. (See Fig. 21)

To the best of our knowledge, the most widely used dataset is NUS, since it has many categories for comparison. Moreover, DeepStab is the most widely used dataset for training in the learning-based method, which contains the most paired videos with the ground truth. However, the color frames are highly similar, training an RGB-based network with this dataset is inherently overfitting and results in the difficulty of generalization. Selfie is proposed for selfie video stabilization and Video + sensor is proposed for 3D video stabilization. These completions are making great contributions to the development of video stabilization and helping develop new methods for various applications.

5.2. Performance of methods

We evaluate them in terms of three metrics (cropping, distortion and stability) on NUS, Selfie, DeepStab and Video + sensor datasets in Table 5. Table 6 reports the comparison of running time

on NUS datasets. We make the following observations regarding the results.

- In the NUS datasets, Hybrid [15], DIFRINT [17] and Deep motion blind [18] achieve the best results from the cropping metrics. Meshflow [10], Hybrid and DIFRINT perform better in the distortion metrics. L1Stabilizer [8] and Bundle [9] achieve the best results in the stability metrics. Among these methods, Hybrid lies in the learning-based hybrid-space fusion. The performance improvement of DIFRINT is achieved by iteratively interpolating. The contribution of Meshflow is based on the proposed sparse and spatially smooth motion field. L1Stabilizer benefits from multiple constraints on the camera path. Bundle achieves better performance due to mesh-based and space-time path optimization.
- In the DeepStab datasets, Hybrid and DIFRINT achieve the best results from the cropping metrics. Hybrid performs better than other methods in the distortion metrics. L1Stabilizer outperforms the others in the stability metrics.
- In the Selfie datasets, Hybrid and DIFRINT also get the best results from the cropping metrics. In the distortion and stability metrics, Deep Selfie VS [20] achieves the best results due to propose 1D linear CNN to estimate the moving least square warp notes. Selfie VS performs better by minimizing the background smoothness in the stability.
- In the Video + sensor datasets, DIFRINT also gets the best results from the cropping metrics. In the distortion metrics, the performance of PW-StableNet [24] is proposed by multi-stage cascade encoder-decoder architectures, which handle distortion better.

Table 5
Quantitative evaluations with the state-of-the-art methods on the NUS, DeepStab, the Selfie and Video + Sensor datasets.

	NUS			DeepStab			Selfie			Video + Sensor		
	C	D	S	C	D	S	C	D	S	C	D	S
Adobe (Subspace) [7]	0.74	0.82	0.87	0.73	0.87	0.83	0.71	0.80	0.84	-	-	-
L1Stabilizer [8]	0.74	0.92	0.89	0.74	0.92	0.85	0.75	0.92	0.85	0.64	0.90	0.82
Bundle [9]	0.84	0.93	0.89	0.76	0.91	0.84	0.68	0.82	0.85	-	-	-
Meshflow [10]	0.55	0.96	0.81	-	-	-	-	-	-	-	-	-
StabNet [11]	0.66	0.88	0.82	0.65	0.86	0.80	0.70	0.78	0.83	0.75	0.85	0.84
STNs [12]	0.82	0.90	0.85	-	-	-	-	-	-	-	-	-
CNN weight space [13]	0.78	0.88	0.84	-	-	-	-	-	-	-	-	-
Deep optical flow [14]	0.86	0.91	0.85	0.82	0.92	0.81	0.78	0.79	0.87	0.79	0.85	0.84
Hybrid [15]	1.00	0.96	0.85	1.00	0.96	0.81	1.00	0.87	0.87	-	-	-
DUT [16]	0.70	0.95	0.83	-	-	-	-	-	-	-	-	-
DIFRINT [17]	1.00	0.96	0.83	1.00	0.94	0.81	1.00	0.87	0.85	1.00	0.88	0.78
Deep motion blind [18]	1.00	0.95	0.81	-	-	-	-	-	-	-	-	-
Selfie VS [19]	-	-	-	-	-	-	0.85	0.91	0.88	-	-	-
Deep Selfie VS [20]	-	-	-	-	-	-	0.88	0.97	0.88	-	-	-
DUT + OVS [21]	0.97	0.93	0.85	-	-	-	-	-	-	-	-	-
Deep3D [22]	0.66	0.92	0.83	-	-	-	-	-	-	-	-	-
Deep-FVS [23]	-	-	-	-	-	-	-	-	-	0.91	0.94	0.85
PW-StableNet [24]	0.88	0.92	0.83	-	-	-	-	-	-	0.93	0.97	0.83
PixStabNet [25]	0.49	0.90	0.83	-	-	-	-	-	-	-	-	-
Self-Supervised [26]	0.95	0.95	0.88	-	-	-	-	-	-	-	-	-

Table 6

Average processing speed (FPS) of different methods on NUS datasets.

Methods	Category	FPS	Future Frames
Adobe (Subspace) [7]	hand-craft	4.2 (CPU)	Yes
L1Stabilizer [8]	hand-craft	10.0 (CPU)	Yes
Bundle [9]	hand-craft	3.5 (CPU)	Yes
MeshFlow [10]	hand-craft	22.0 (CPU)	No
StabNet [11]	CNN	35.5 (GPU)	No
STNs [12]	CNN	30.1 (GPU)	No
CNN weight space [13]	CNN	0.2(GPU)	Yes
Deep optical flow [14]	CNN	1.8(GPU)	Yes
Hybrid [15]	CNN	2.6(GPU)	Yes
DUT [16]	CNN	14 (GPU)	Yes
DIFRINT [17]	CNN	14.3 (GPU)	Yes
Deep3D [22]	CNN	34.5 (GPU)	No
PWStableNet [24]	CNN	56.2 (GPU)	Yes
PixStableNet [25]	CNN	54.6 (GPU)	No
Self-Supervised [26]	CNN	41(GPU)	No

In the stability metrics, Deep-FVS [23] achieves the best results, which demonstrates the effectiveness of the fusion of sensor data and optical flow.

- In general, learning-based methods outperform the conventional methods across all datasets in the cropping and distortion metrics. This is likely due to CNN can extract intra-frame feature information better than traditional methods. In the stability metrics, conventional methods perform better than the learning-based methods on the NUS and DeepStab datasets. In contract, learning-based methods achieve better performance on the Selfie, Video + sensor datasets. This phenomenon indicates that hand-craft based methods can better extra inter-frame motion information. However, CNN is designed for specific scenes, such as Selfie and 3D stabilization, which is proved to be effective.

In Table 6, Subspace [7] has a speed of 4 fps, and the major time-consuming components are KLT tracker and content-preserving warping. L1Stabilizer [8] has a speed of 10.0 fps because it is based on per-frame LK-tracker only without 3D reconstruction. Bundle [9] has a speed of 3.5 fps, mainly because it spends 300 ms to extract SURF features. MeshFlow [10] achieves a speed of 22 fps because it uses vertex profiles to well approximate long feature tracks which reduces the computing costs. StabNet [11] achieves a speed of 35.5 fps because it avoids explicitly computing camera paths but instead uses a CNN model to predict warping functions directly. STNs [12] achieves a speed of 26 fps because it converts the video stabilization problems to an affine transformation generation between adjacent frames, avoiding heavy computing. CNN weight space [13] takes 6 s to process per frame because it uses optical flow to track and optimize in the neural network, which is time-consuming. Deep optical flow [14] has a speed of 1.8 fps because it takes 270 ms to extract SURF features and spends 300 ms to stabilize the frames. Hybrid [15] spends 3.8 s to process each frame, mainly because motion smoothing [14] takes about 2.6 s. DUT [16] achieves a speed of 14 fps because it explicitly estimates and smooths trajectories using unsupervised deep learning. DIFRINT [17] achieves a speed of 14.3 fps and it consists of optical flow estimators, warping layers and feed forwarding. Since its network involves 2D convolutions, it is an offline method requiring future frames and multiple iterations. Deep3D [22] achieves a speed of 34.5 fps because it employs parameter-propagation to give more than doubled speed in the frame rectification stage. PWStableNet [24] achieves a speed of 56.2 fps because it is constructed with three-stage cascaded encoder-decoder modules. However, the use of future frames causes a delay of several frames. PixStableNet [25] achieves a speed of 54.6 fps without future frames, owing to its proposed multi-scale CNN. Self-Supervised

[26] achieves a speed of 41 fps because it consists of simple and efficient modules. In summary, learning-based methods are faster than conventional methods on NUS datasets in terms of FPS. On the one hand, the result may be caused by the difference between CPU and GPU. On the other hand, most conventional methods are based on feature matching for motion estimation, which is time-consuming.

6. Challenges and future directions

Although rapid development and promising progress have been achieved by state-of-the-art video stabilization methods, especially on some public benchmark datasets, there are many challenges and open problems. We discuss current challenges and future research directions below.

6.1. Current challenges

Lack of recognized assessment metrics. At present, a widely used assessment metric is proposed by [9], it just considers cropping, distortion, stability metrics but ignores other indicators. Due to the lack of comprehensive and fair assessment metrics, it is impossible to evaluate the performance objectively. So, evaluating the stabilization quality appears as a hot and challenging research topic.

Lack of benchmark data. Despite several public datasets being proposed, there is still not a large dataset that contains enough unsteady and steady videos. The capacity of datasets limits the generalization of learning-based methods. Furthermore, the lack of benchmark data makes the method assessment biased in comparing the performance.

Stability problems. Stability is always a critical challenge, many algorithms are proposed to solve it. However, current SOTA methods achieve unsatisfactory results. On one hand, traditional method only utilizes simple linear transformation to fit camera motion path. On the other hand, CNN cannot extra motion feature directly in inter-frame information, which causes the CNN methods to have poor performance in stability.

Missing image pixels, moving occlusion and parallax problems. Videos miss frame boundary pixels after motion compensation which lowers the video quality. Stabilizing video with large camera motion and moving occlusion is challenging, their results are thus prone to severe artifacts and distorted. Video with non-trivial depth variation in the scene is challenging, 2D linear model is weak for the parallax scenes, while 3D model can achieve high-quality results but are more fragile.

Variety of scenes. With the development of smart devices, video stabilization can be widely used in many scenes, such as selfie and UAVs. The stabilization requirements in different scenes are not the same. For instance, there is always non-rigid foreground motion in selfie videos, which is different from other scenes.

Real-time video stabilization. Real-time is a challenging problem in video stabilization. Warping-based methods adopt feature matching in motion estimation, which leads to slow running speed. With the development of deep learning, many CNN-based methods are proposed and the running speed reaches real-time. However, these methods use future frames as network input, which leads to several frame delays.

6.2. Future directions

In this section, future research directions are discussed.

Assessment metrics. Exploiting more perceptual attributes is very important to build reasonable mathematical models. Current widely metrics only involve three indicators. To address this issue,

future work should explore the illumination conditions, the characteristics of the vision systems, the accurate recognition of background area and more characteristics to analyze their internal connections.

Benchmark data. There is not a large dataset that contains enough paired videos now. Future work should explore to build a large dataset; however, it is time-consuming. Furthermore, there are still several potential approaches that have not been explored, such as the semi-supervised learning, which is designed to address the data problems.

Effectiveness of video stabilization. Video stabilization is a hot research topic. On the one hand, future work should design the network architecture to extract inter-frame motion features for reconstructing the trajectory, improving stabilization effectiveness. On the other hand, the fusion between conventional and learning-based methods should be explored to utilize the advantages of hand-craft and CNN methods for video stabilization.

Stabilized video quality (cropping, distortion and parallax). In order to achieve full-frame video stabilization, interpolating-based and hybrid-based methods are proposed. However, their results both bring artifacts and distortion. Future work should explore to combine the sensor dates and frame content, which models the virtual camera pose and sensor dates for correcting the pose. In this way, motion rigidity is naturally preserved without artifacts. Because it achieves more accurate motion inference to obtain better stable results without large cropping, at the same time, it can handle parallax owing to extra sensor data.

Video stabilization in specific scenes. General video stabilization has its own limitation in different scenes, because it ignores scene characteristics. The Selfie datasets results have proved it. So, future work should design video stabilization according to the specific scenes.

Real-time video stabilization. Real-time implementation is a critical challenge. Conventional methods are mainly offline, and owing to that it takes much time for feature matching to reconstruct motion trajectories. Meshflow uses vertex profiles to replace the feature matching which reduces computation. However, it misses many frame boundary pixels after motion compensation. CNN methods achieve real-time due to their lightweight and efficient architecture; however, some methods lead to several frame delays. Therefore, reconstructing feature tracks without feature matching is a perspective direction in conventional methods. In learning-based methods, the network design without frame delays is also future work.

7. Conclusion

This paper presents a comprehensive survey of video stabilization. In particular, we divide various methods into conventional and deep learning-based methods. Moreover, we discuss the video stabilization quality assessment. We also make discussion on the results demonstrated by state-of-the-art methods. Although the results are promising, there remain potential and unsolved challenges. Finally, We further discuss several current challenges and future directions to gain thorough understandings. We hope this survey can give readers an understanding of video stabilization comprehensively.

CRediT authorship contribution statement

Yiming Wang: Methodology, Investigation, Data curation, Writing – original draft. **Qian Huang:** Conceptualization, Writing – review & editing, Supervision. **Chuanxu Jiang:** Software, Visualization. **Jiwen Liu:** Validation. **Mingzhou Shang:** Resources. **Zhuang Miao:** Conceptualization, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Yiming Wang received the B.Eng.degree from Jiangsu Ocean university, China, in 2016, and the M.Sc.degree from Hohai university, in 2019. From 2019 to 2021, he worked in Jiangsu Telecom Company Limited, Nanjing, China. Since Sep.2021, He is pursuing the Ph.D. degree with the Computer Science and Technology in Hohai university, Nanjing, China. His current research interests include video stabilization and video compression.



Jiwen Liu was born in Xinghua, China, in 1999. He received the B.Eng.degree from the Jinling Institute of Technology, China, in 2021. He is currently pursuing his master's degree in Hohai University. His research interests include image processing and video stabilization.



Qian Huang received the B.Sc.degree in computer science from Nanjing University, China, in 2003, and the Ph.D.degree in computer science from the Institute of Computing Technology, Chinese Academy of Sciences, in 2010. From 2010 to 2012, he was a deputy technical manager of Mediatek (Beijing) Incorporation, Beijing, China. Since Dec. 2012, he is with Hohai University, Nanjing, China, where he serves as the dean of Computer Science & Technology Department. His research interests include media computing, data mining, and intelligent education. He is a member of the Chinese Association for Artificial Intelligence Technical Committee on Deep Learning, a member of the China Computer Federation Technical Committee on Multimedia Technology, and a member of the China Society of Image and Graphics Technical Committee on Multimedia. He holds 2 US patents and has published 6 books as well as more than 40 technical articles.



Mingzhou Shang was born in Xuzhou, China, in 1998. He received the B.Eng.degree from Hohai University, Nanjing, China, in 2020. He is currently pursuing his master degree in Hohai University. His research interests involve computer vision and deep learning.



Chuanxu Jiang was born in Qingdao, China, in 1998. He received the B.Eng. degree from the Qufu normal University, China in 2021. Now is the postgraduate student in HoHai University, China. His research interests include video stabilization and computer vision.



Zhuang Miao received the Ph.D. degree from the PLA University of Science and Technology, Nanjing, China, in 2007. He is currently a Professor with the Army Engineering University of PLA, Nanjing. His current research interests include artificial intelligence, pattern recognition, and computer vision.